A First Course in Abstract Algebra

Part 10: Automorphisms and Galois Theory

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

Contents

48 Automorphisms of Fields 7

Exercise 48.1: Conjugates of a square root 7

Exercise 48.2: Changing the base field 7

Exercise 48.3: A translated square root 7

Exercise 48.4: A difference of square roots 7

Exercise 48.5: A mixed real and imaginary generator 7

Exercise 48.6: Conjugacy over the real field 7

Exercise 48.7: A nested square root over the rationals 7

Exercise 48.8: A nested square root over a quadratic base 8

Exercise 48.9: A radical fixed by a sign change 8

Exercise 48.10: Applying a sign change to a sum 8

Exercise 48.11: Composing two sign changes 8

Exercise 48.12: An automorphism of a quotient 8

Exercise 48.13: Simplifying a power of an automorphism 8

Exercise 48.14: A nested composition 8

Exercise 48.15: Fixed fields in a biquadratic extension 8

Exercise 48.16: The fixed field of one radical flip . . . 9
Exercise 48.17: The square of a radical flip . . . 9
Exercise 48.18: Fixing two separate sign changes . . . 9
Exercise 48.19: Fixing two simultaneous sign changes . . . 9
Exercise 48.20: Fixing all three simultaneous signs . . . 9
Exercise 48.21: Fixing each of the three generators . . . 9
Exercise 48.22: The eight-element sign-change group . . . 9
Exercise 48.23: Correcting the conjugate criterion . . . 10
Exercise 48.24: The kernel criterion for conjugacy . . . 10
Exercise 48.25: A shifted generator gives the same automorphism . . . 10
Exercise 48.26: Frobenius on four elements . . . 10
Exercise 48.27: Frobenius on nine elements . . . 11
Exercise 48.28: Frobenius on an infinite field . . . 11
Exercise 48.29: Ten statements about field automorphisms . . . 11
Exercise 48.30: Synopsis of the existence direction . . . 11
Exercise 48.31: Synopsis of the necessity direction . . . 11
Exercise 48.32: Bounding the number of embeddings . . . 12
Exercise 48.33: Images of generators determine a map . . . 12
Exercise 48.34: Permuting the conjugates present in a field . . . 12
Exercise 48.35: A generating set has the same fixed field as its group . . . 12
Exercise 48.36: Prime cyclotomic fields . . . 12
Exercise 48.37: Transcendental generators . . . 12
Exercise 48.38: All automorphisms of a rational function field . . . 13
Exercise 48.39: Rigidity of real-field automorphisms . . . 13
49 The Isomorphism Extension Theorem . . . 14
Exercise 49.1: Extending an identity on two radicals . . . 14
Exercise 49.2: Extending a prescribed negative product . . . 14
Exercise 49.3: Extending conjugation on a triple product . . . 14
Exercise 49.4: Embeddings of a real cubic field . . . 14
Exercise 49.5: Adding an independent square root . . . 14
Exercise 49.6: Extending a prescribed quadratic conjugation . . . 14
Exercise 49.7: A transcendental sign change . . . 15
Exercise 49.8: Ten statements about extensions of isomorphisms . . . 15
Exercise 49.9: An algebraic self-embedding of a closed field . . . 15

Exercise 49.10: Extending to an automorphism of the closure . . . 15
Exercise 49.11: Closures of algebraically related fields . . . 16
Exercise 49.12: Closures of rational function fields . . . 16
Exercise 49.13: Index is bounded by degree . . . 16
50 Splitting Fields . . . 17
Exercise 50.1: Splitting a quadratic . . . 17
Exercise 50.2: Fourth roots of unity . . . 17
Exercise 50.3: Two distinct quadratic radicals . . . 17
Exercise 50.4: Splitting an irreducible cubic . . . 17
Exercise 50.5: Third roots of unity . . . 17
Exercise 50.6: Combining square and cube roots . . . 17
Exercise 50.7: Automorphisms of the real cubic field . . . 17
Exercise 50.8: Automorphisms of its splitting field . . . 18
Exercise 50.9: Fixing the real cubic generator . . . 18
Exercise 50.10: A cubic splits in an eight-element field . . . 18
Exercise 50.11: Correcting the splitting-field criterion . . . 18
Exercise 50.12: Correcting the meaning of splitting . . . 18
Exercise 50.13: Bounds for a splitting-field degree . . . 18
Exercise 50.14: Ten statements about splitting fields . . . 18
Exercise 50.15: Why irreducibility is essential . . . 19
Exercise 50.16: Group orders in a tower . . . 19
Exercise 50.17: One polynomial suffices for a finite normal extension . . . 19
Exercise 50.18: A quadratic extension is a splitting field . . . 19
Exercise 50.19: Containing all conjugates . . . 20
Exercise 50.20: No nontrivial automorphism of the real cubic . . . 20
Exercise 50.21: The cyclic subgroup fixing the quadratic radical . . . 20
Exercise 50.22: The permutation representation on roots . . . 20
Exercise 50.23: The full cubic permutation group . . . 20
Exercise 50.24: A prime-root-of-unity extension . . . 20
Exercise 50.25: Uniqueness of splitting fields in different closures . . . 21
51 Separable Extensions . . . 22
Exercise 51.1: Combining square and cube roots . . . 22
Exercise 51.2: Combining fourth and sixth roots . . . 22
Exercise 51.3: A primitive biquadratic generator . . . 22

Exercise 51.4: A primitive generator for a complex sextic field . . . 22
Exercise 51.5: Where multiplicity is measured . . . 23
Exercise 51.6: The definition of separable element . . . 23
Exercise 51.7: A repeated polynomial over a perfect field . . . 23
Exercise 51.8: Ten statements about separability . . . 23
Exercise 51.9: Separable elements are closed under field operations . . . 23
Exercise 51.10: A basis for a Frobenius extension . . . 24
Exercise 51.11: Algebraic extensions of a perfect field . . . 24
Exercise 51.12: Transitivity for infinite separable extensions . . . 24
Exercise 51.13: The separable closure inside an algebraic extension . . . 24
Exercise 51.14: Frobenius generates a finite-field Galois group . . . 24
Exercise 51.15: The additive derivative homomorphism . . . 24
Exercise 51.16: Formal derivative rules . . . 25
Exercise 51.17: Multiplicity and vanishing derivative . . . 25
Exercise 51.18: Characteristic zero forces irreducible separability . . . 25
Exercise 51.19: The derivative test for an inseparable irreducible . . . 25
Exercise 51.20: Common factors over the algebraic closure . . . 25
Exercise 51.21: Bringing the gcd test back to the coefficient field . . . 26
Exercise 51.22: A root-free computational test . . . 26
52 Totally Inseparable Extensions . . . 27
Exercise 52.1: A separable closure with two independent exponents . . . 27
Exercise 52.2: A separable closure with a mixed monomial . . . 27
Exercise 52.3: A totally inseparable closure in the independent case . . . 27
Exercise 52.4: A totally inseparable closure in the mixed case . . . 27
Exercise 52.5: Ten statements about totally inseparable extensions . . . 28
Exercise 52.6: Purely inseparable elements form a field . . . 28
Exercise 52.7: Perfection and surjective Frobenius . . . 28
Exercise 52.8: Perfection through a finite extension . . . 29
53 Galois Theory . . . 30
Exercise 53.1: The embedding index of a multiquadratic field . . . 30
Exercise 53.2: The group order of a multiquadratic field . . . 30
Exercise 53.3: The subgroup fixing the bottom field . . . 30
Exercise 53.4: Fixing two specified radicals . . . 30
Exercise 53.5: Fixing a product radical . . . 30

Exercise 53.6: Fixing a triple-product radical . . . 30
Exercise 53.7: Fixing a sum of two radicals . . . 30
Exercise 53.8: Fixing the entire top field . . . 30
Exercise 53.9: The group of fourth roots of unity . . . 31
Exercise 53.10: A relative finite-field Frobenius . . . 31
Exercise 53.11: The cubic splitting-field correspondence . . . 31
Exercise 53.12: The group of two quadratic factors . . . 32
Exercise 53.13: The group of third roots of unity . . . 32
Exercise 53.14: An abstract group does not determine its field . . . 32
Exercise 53.15: Ten statements about the Galois correspondence . . . 32
Exercise 53.16: Abelian Galois extensions in a tower . . . 33
Exercise 53.17: Norm and trace lie in the base field . . . 33
Exercise 53.18: Eight norm and trace calculations . . . 33
Exercise 53.19: Norm and trace from minimal-polynomial coefficients . . . 34
Exercise 53.20: The group order divides the factorial . . . 34
Exercise 53.21: The natural permutation group of a polynomial . . . 34
Exercise 53.22: Cyclotomic extensions over a general field . . . 34
Exercise 53.23: Cyclic extensions and their intermediate fields . . . 34
Exercise 53.24: The polynomial formed from every automorphism . . . 35
Exercise 53.25: A compositum corresponds to an intersection . . . 35
Exercise 53.26: An intersection corresponds to a generated subgroup . . . 35

54 Illustrations of Galois Theory . . . 36

Exercise 54.1: Irreducibility of the eighth cyclotomic polynomial . . . 36
Exercise 54.2: Checking the entire octic fixed-field diagram . . . 36
Exercise 54.3: Primitive generators and all their minimal polynomials . . . 37
Exercise 54.4: The fifth-root-of-unity correspondence . . . 37
Exercise 54.5: A fifth radical over a cyclotomic base . . . 38
Exercise 54.6: The seventh-root-of-unity correspondence . . . 38
Exercise 54.7: Eighth roots without recomputing a table . . . 38
Exercise 54.8: A complete quartic splitting-field correspondence . . . 38
Exercise 54.9: Two symmetric-function identities . . . 40
Exercise 54.10: Polynomials of a sum and of squares . . . 40
Exercise 54.11: Every finite group occurs over a suitable field . . . 41
Exercise 54.12: Discriminants and even permutations . . . 41

Exercise 54.13: Algebraic integers form a ring 41
55 Cyclotomic Extensions 42
Exercise 55.1: Multiplying the primitive eighth-root factors 42
Exercise 55.2: The twentieth-root group as an abelian group 42
Exercise 55.3: Three Euler totients 42
Exercise 55.4: The first thirty constructible regular polygons 42
Exercise 55.5: The smallest constructible integral-degree angle 43
Exercise 55.6: Twelfth roots and a group of exponent two 43
Exercise 55.7: Two cyclotomic polynomials in finite characteristic 43
Exercise 55.8: A repeated polynomial already split over the prime field 43
Exercise 55.9: Ten statements about cyclotomic fields 43
Exercise 55.10: Partitioning roots by their orders 44
Exercise 55.11: The first six cyclotomic polynomials 44
Exercise 55.12: The twelfth cyclotomic polynomial 44
Exercise 55.13: Negating roots of odd order 44
Exercise 55.14: The compositum for coprime root orders 45
Exercise 55.15: The product decomposition for coprime orders 45
56 Insolvability of the Quintic 46
Exercise 56.1: A radical extension in characteristic two 46
Exercise 56.2: An even polynomial of degree eight 46
Exercise 56.3: Ten statements about radicals and solvability 46
Exercise 56.4: The quadratic formula over a field 46
Exercise 56.5: A biquadratic solved by successive radicals 47
Exercise 56.6: Refining a series with abelian factors 47
Exercise 56.7: Refining solvable factors to abelian composition factors 47
Exercise 56.8: An explicit quintic with no radical solution 47

48 Automorphisms of Fields

Exercise 48.1: Conjugates of a square root

The minimal polynomial over $\mathbb{Q}$ is $x^2 - 2$, irreducible by Eisenstein at 2. Its two zeros are $\sqrt{2}, -\sqrt{2}$, so these are exactly the conjugates.

Exercise 48.2: Changing the base field

Since $\sqrt{2} \in \mathbb{R}$, its minimal polynomial over $\mathbb{R}$ is $x - \sqrt{2}$. Its only conjugate over $\mathbb{R}$ is itself. Conjugacy always depends on the base field.

Exercise 48.3: A translated square root

Subtracting 3 gives $(x-3)^2 - 2 = x^2 - 6x + 7$. This is irreducible over $\mathbb{Q}$ because a rational zero would make $\sqrt{2}$ rational. Its roots, hence all conjugates, are $3 + \sqrt{2}$ and $3 - \sqrt{2}$.

Exercise 48.4: A difference of square roots

Independent changes of signs of $\sqrt{2}$ and $\sqrt{3}$ give $\pm\sqrt{2} \pm \sqrt{3}$. They are four distinct conjugates of $\sqrt{2} - \sqrt{3}$. Squaring twice gives $x^4 - 10x^2 + 1 = 0$. To see the degree is 4, put $\alpha = \sqrt{2} - \sqrt{3}$. Since $\alpha^{-1} = -(\sqrt{2} + \sqrt{3})$, both square roots lie in $\mathbb{Q}(\alpha)$. Thus this field is the degree-four field $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, and the list is complete.

Exercise 48.5: A mixed real and imaginary generator

The four sign choices $\pm\sqrt{2} \pm i$ are the images under the four automorphisms of $\mathbb{Q}(\sqrt{2}, i)$. They are distinct. The field has degree 4 because i is not in the real field $\mathbb{Q}(\sqrt{2})$. Thus the minimal polynomial is $x^4 - 2x^2 + 9$, obtained by multiplying the four linear factors, and the four listed numbers are all conjugates.

Exercise 48.6: Conjugacy over the real field

Over $\mathbb{R}$, the minimal polynomial is $(x - \sqrt{2})^2 + 1$, which has no real zero. The conjugates are therefore $\sqrt{2} + i$ and $\sqrt{2} - i$. The base field fixes $\sqrt{2}$, so its sign cannot change.

Exercise 48.7: A nested square root over the rationals

Let $\alpha = \sqrt{1 + \sqrt{2}}$. Its square gives $\sqrt{2} = \alpha^2 - 1$, so $\mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\alpha)$. The element $1 + \sqrt{2}$ is not a square in $\mathbb{Q}(\sqrt{2})$: its norm to $\mathbb{Q}$ is -1, whereas the norm of a square is a rational square. Hence $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$. Eliminating the inner root gives $x^4 - 2x^2 - 1$. Its four zeros are $\pm\sqrt{1 + \sqrt{2}}$ and $\pm\sqrt{1 - \sqrt{2}}$, with the latter square root taken in $\mathbb{C}$. These are all conjugates.

Exercise 48.8: A nested square root over a quadratic base

Over $\mathbb{Q}(\sqrt{2})$, the polynomial is $x^2 - (1 + \sqrt{2})$. The norm argument in Exercise 48.7 shows it is irreducible. Thus only the two outer signs can change: the conjugates are $\sqrt{1 + \sqrt{2}}$ and $-\sqrt{1 + \sqrt{2}}$.

Exercise 48.9: A radical fixed by a sign change

The automorphism τ_2 changes only the sign of $\sqrt{2}$. Therefore $\tau_2(\sqrt{3}) = \sqrt{3}$.

Exercise 48.10: Applying a sign change to a sum

An automorphism preserves addition and fixes the other generating radicals. Thus $\tau_2(\sqrt{2} + \sqrt{5}) = -\sqrt{2} + \sqrt{5}$.

Exercise 48.11: Composing two sign changes

Apply the rightmost map first. The map τ_2 sends $\sqrt{2} + 3\sqrt{5}$ to $-\sqrt{2} + 3\sqrt{5}$; τ_3 fixes both radicals appearing there. Hence $(\tau_3\tau_2)(\sqrt{2} + 3\sqrt{5}) = -\sqrt{2} + 3\sqrt{5}$.

Exercise 48.12: An automorphism of a quotient

The map $\tau_5\tau_3$ flips $\sqrt{5}$ and $\sqrt{3}$ and fixes $\sqrt{2}$. Because field automorphisms preserve division,

$$(\tau_5\tau_3)\left(\frac{\sqrt{2} - 3\sqrt{5}}{2\sqrt{3} - \sqrt{2}}\right) = \frac{\sqrt{2} + 3\sqrt{5}}{-2\sqrt{3} - \sqrt{2}}.$$

The denominator remains nonzero because an injective map cannot send a nonzero element to zero.

Exercise 48.13: Simplifying a power of an automorphism

Since τ_5^2 is the identity and $\sqrt{45} = 3\sqrt{5}$, the remaining $\tau_3\tau_2$ flips only the $\sqrt{2}$ term. The result is $-\sqrt{2} + \sqrt{45}$.

Exercise 48.14: A nested composition

First $(\tau_2\tau_5)(\sqrt{30}) = \sqrt{30}$ because two radical signs change. Applying τ_5 to the bracket gives $\sqrt{2} - \sqrt{3} - \sqrt{30}$. Finally τ_3 changes the signs of the last two radical terms. The answer is $\sqrt{2} + \sqrt{3} + \sqrt{30}$.

Exercise 48.15: Fixed fields in a biquadratic extension

Write an element uniquely as $a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}$ with rational coefficients. In the source notation σ_1 flips $\sqrt{2}$, σ_2 flips $\sqrt{3}$, and σ_3 flips both.

(a) Fixing both σ_1 and σ_3 forces $b = c = d = 0$, so the fixed field is $\mathbb{Q}$.

(b) Fixing σ_3 forces $b = c = 0$, leaving $\mathbb{Q}(\sqrt{6})$.

(c) Fixing both σ_2 and σ_3 again forces $b = c = d = 0$, giving $\mathbb{Q}$.

Exercise 48.16: The fixed field of one radical flip

Use the basis $1, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{10}, \sqrt{15}, \sqrt{30}$ of E. The map τ_3 negates exactly those basis terms containing $\sqrt{3}$. Their coefficients must vanish in a fixed element. The surviving terms span $\mathbb{Q}(\sqrt{2}, \sqrt{5})$, which is the fixed field.

Exercise 48.17: The square of a radical flip

The map τ_3^2 is the identity, so every element is fixed. Its fixed field is $E = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$.

Exercise 48.18: Fixing two separate sign changes

An element fixed by both τ_2 and τ_3 has no basis term involving $\sqrt{2}$ or $\sqrt{3}$. Only $1, \sqrt{5}$ survive, so the fixed field is $\mathbb{Q}(\sqrt{5})$.

Exercise 48.19: Fixing two simultaneous sign changes

The map $\tau_5\tau_2$ negates $\sqrt{2}, \sqrt{5}$ and fixes $\sqrt{3}$. A basis monomial survives exactly when it contains both or neither of $\sqrt{2}, \sqrt{5}$. The surviving basis is $1, \sqrt{3}, \sqrt{10}, \sqrt{30}$, so the fixed field is $\mathbb{Q}(\sqrt{3}, \sqrt{10})$.

Exercise 48.20: Fixing all three simultaneous signs

A basis monomial is fixed by $\tau_5\tau_3\tau_2$ exactly when it contains an even number of the three radicals. Thus the fixed field has basis $1, \sqrt{6}, \sqrt{10}, \sqrt{15}$ and equals $\mathbb{Q}(\sqrt{6}, \sqrt{10})$, since $\sqrt{6}\sqrt{10} = 2\sqrt{15}$.

Exercise 48.21: Fixing each of the three generators

Every nonconstant basis monomial contains at least one generating radical and is negated by its sign flip. Requiring invariance under all three flips leaves just rational multiples of 1. The fixed field is $\mathbb{Q}$.

Exercise 48.22: The eight-element sign-change group

(a) Each τ_j is nonidentity, since it negates a nonzero radical, and its square fixes all generators. It therefore has order 2.

(b) Put $a = \tau_2, b = \tau_3, c = \tau_5$. They commute because independent sign changes commute on each

generator. The elements are $1, a, b, c, ab, ac, bc, abc$, with table

	1	a	b	c	ab	ac	bc	abc
1	1	a	b	c	ab	ac	bc	abc
a	a	1	ab	ac	b	c	abc	bc
b	b	ab	1	bc	a	abc	c	ac
c	c	ac	bc	1	abc	a	b	ab
ab	ab	b	a	abc	1	bc	ac	c
ac	ac	c	abc	a	bc	1	ab	b
bc	bc	abc	c	b	ac	ab	1	a
abc	abc	bc	ac	ab	c	b	a	1

The rule is to cancel repeated letters, since $a^2 = b^2 = c^2 = 1$.

(c) Any rational automorphism sends each $\sqrt{j}$ to $\pm\sqrt{j}$ because it preserves its minimal polynomial. There are at most $2^3 = 8$ choices. The eight maps above realize every choice, so they form the full group, isomorphic to $\mathbb{Z}_2^3$.

Exercise 48.23: Correcting the conjugate criterion

Replace "the same polynomial" by "the same irreducible polynomial over F." Equivalently, α, β have the same monic minimal polynomial over F. Merely being roots of one polynomial is insufficient: 0 and 1 are both roots of $x(x-1)$, but have different minimal polynomials.

Exercise 48.24: The kernel criterion for conjugacy

The statement is correct. The evaluation kernel at an algebraic element is the principal ideal generated by its monic minimal polynomial. Equality of the kernels is equivalent to equality of their monic generators, hence to conjugacy over F.

Exercise 48.25: A shifted generator gives the same automorphism

(a) The other conjugate of $\alpha = 3 + \sqrt{2}$ is $\beta = 3 - \sqrt{2}$, since both satisfy $x^2 - 6x + 7$.

(b) Both conjugation automorphisms fix $\mathbb{Q}$ and send $\sqrt{2} = \alpha - 3$ to $-\sqrt{2} = \beta - 3$. Since $\sqrt{2}$ generates the field, they are exactly the same map; only the chosen generator used to describe it differs.

Exercise 48.26: Frobenius on four elements

Write the field as $\mathbb{Z}_2(\alpha)$ with $\alpha^2 + \alpha + 1 = 0$, so $\alpha^2 = \alpha + 1$. Squaring gives

$$0 \mapsto 0, \quad 1 \mapsto 1, \quad \alpha \mapsto \alpha + 1, \quad \alpha + 1 \mapsto \alpha.$$

Thus precisely 0 and 1 are fixed; the fixed field is $\mathbb{Z}_2$.

Exercise 48.27: Frobenius on nine elements

Here $\alpha^2 = -1 = 2$ in characteristic 3. For $a, b \in \mathbb{Z}_3$, cubing gives $(a + b\alpha)^3 = a - b\alpha$. Thus 0,1,2 are fixed; the other six elements are exchanged in the pairs

$$\alpha \leftrightarrow 2\alpha, \quad 1 + \alpha \leftrightarrow 1 + 2\alpha, \quad 2 + \alpha \leftrightarrow 2 + 2\alpha.$$

The fixed field is $\mathbb{Z}_3$.

Exercise 48.28: Frobenius on an infinite field

Let $F = \mathbb{Z}_p(t)$ with t transcendental. The image of $r(t) \mapsto r(t)^p$ is $\mathbb{Z}_p(t^p)$. It does not contain t: if $(P(t)/Q(t))^p = t$ with P, Q nonzero polynomials, comparing degrees in $P(t)^p = tQ(t)^p$ gives $p \deg P = 1 + p \deg Q$, impossible modulo p. Frobenius remains injective, but need not be onto.

Exercise 48.29: Ten statements about field automorphisms

(a) False. An automorphism fixes 0 and 1 and cannot map one to the other.

(b) False. $\mathbb{Q}(0) = \mathbb{Q}$ and $\mathbb{Q}(\sqrt{2})$ are not isomorphic: any isomorphism fixes $\mathbb{Q}$, and their rational degrees differ.

(c) True, by the conjugation isomorphism theorem.

(d) True. It fixes 1, hence integers and their quotients, giving the prime field.

(e) False. The identity automorphism of a finite field fixes only finitely many elements.

(f) True: it fixes the distinct elements 0 and 1.

(g) True: it fixes the infinite prime subfield $\mathbb{Q}$.

(h) True: composition, identity, and inverse preserve field operations.

(i) True: fixed elements are closed under differences, products and inverses of nonzero elements, and include 0,1.

(j) True. Fixing the larger field E implies fixing F, and these maps form a subgroup.

Exercise 48.30: Synopsis of the existence direction

When α and β have the same minimal polynomial p, identifying both $F(\alpha)$ and $F(\beta)$ with $F[x]/(p)$ gives the unique isomorphism fixing F and sending α to β. This works because evaluation at either root has exactly the kernel (p).

Exercise 48.31: Synopsis of the necessity direction

An isomorphism fixing the coefficients in F carries the equation $p(\alpha) = 0$ to $p(\beta) = 0$, so the minimal polynomials of α and β coincide. Applying the inverse gives the reverse divisibility if one spells this out via minimal polynomials.

Exercise 48.32: Bounding the number of embeddings

An embedding fixing F must send α to a root of its degree-n minimal polynomial. There are at most n such distinct roots in an algebraic closure. Its image on α determines the embedding on every polynomial in α, hence on all $F(\alpha)$. Therefore there are at most n embeddings.

Exercise 48.33: Images of generators determine a map

Every element of $F(\alpha_1, \ldots, \alpha_n)$ is a rational expression in those generators with coefficients in F. A homomorphism fixing F evaluates that expression at $\sigma(\alpha_1), \ldots, \sigma(\alpha_n)$. Its denominator stays nonzero by injectivity. Thus two such automorphisms agreeing on the generators agree on every field element.

Exercise 48.34: Permuting the conjugates present in a field

Let p be the minimal polynomial of α. If $\beta \in E$ satisfies $p(\beta) = 0$, then $p(\sigma(\beta)) = \sigma(p(\beta)) = 0$, because σ fixes every coefficient of p. So σ carries the set of such roots into itself. Its inverse has the same property, proving the restriction is a bijection, hence a permutation.

Exercise 48.35: A generating set has the same fixed field as its group

Since $S \subseteq H$, any element fixed by H is fixed by S. Conversely, if x is fixed by every $s \in S$, it is fixed by s^{-1} too: apply s^{-1} to $s(x) = x$. It is then fixed by any finite composition of generators and their inverses. Those compositions comprise H, so $x \in E_H$. Thus $E_S = E_H$.

Exercise 48.36: Prime cyclotomic fields

(a) A zero ζ of Φ_p is a nonidentity pth root of unity. Since p is prime, it has order p, and $\zeta, \ldots, \zeta^{p-1}$ are distinct, nonidentity roots. They exhaust the $p - 1$ zeros of Φ_p.

(b) Each choice $\sigma_a(\zeta) = \zeta^a$, $1 \leq a < p$, gives a rational embedding onto the same field because a is invertible modulo p. Composition satisfies $\sigma_a \sigma_b = \sigma_{ab}$, proving the group is abelian and has order $p - 1$.

(c) Multiplication by ζ carries the usual basis $1, \ldots, \zeta^{p-2}$ to the basis $\zeta, \ldots, \zeta^{p-1}$. The automorphisms permute its basis vectors transitively. If $\sum_{j=1}^{p-1} c_j \zeta^j$ is fixed by every map, uniqueness of basis coefficients forces all c_j equal, say to c. The sum is then $c(\zeta + \cdots + \zeta^{p-1}) = -c \in \mathbb{Q}$. Conversely rationals are fixed, so the fixed field is $\mathbb{Q}$.

Exercise 48.37: Transcendental generators

Yes. Define $P(\alpha)/Q(\alpha) \mapsto P(\beta)/Q(\beta)$ for $P, Q \in F[x]$, $Q \neq 0$. Transcendence ensures neither evaluation denominator vanishes, and equality of rational expressions is exactly a polynomial identity, so the map is well defined. It preserves both operations and has inverse obtained by exchanging α, β. Thus it is an isomorphism fixing F.

Exercise 48.38: All automorphisms of a rational function field

They are precisely

$$x \longmapsto \frac{ax+b}{cx+d}, \qquad a,b,c,d \in F, \quad ad-bc \neq 0.$$

Such a substitution has inverse $x \mapsto (dx-b)/(-cx+a)$, so it is an automorphism. Multiplying all four coefficients by the same nonzero scalar gives the same map. For completeness, if an automorphism sends x to $r = P(x)/Q(x)$ in lowest terms, put $m = \max(\deg P, \deg Q)$. The standard degree computation gives $[F(x) : F(r)] = m$: $P(T) - rQ(T)$ is the minimal polynomial of x over $F(r)$. Indeed, with r replaced by an independent indeterminate u, the polynomial $P(T) - uQ(T)$ is irreducible in $F[T,u]$, since a factorization of this degree-one polynomial in u would require a nonconstant common divisor of P, Q; Gauss's lemma transfers this to $F(u)[T]$. An automorphism has $F(r) = F(x)$, so $m = 1$. The numerator and denominator are therefore linear or constant, and coprimality/nonconstancy is exactly $ad - bc \neq 0$.

Exercise 48.39: Rigidity of real-field automorphisms

(a) If $u = v^2$, then $\sigma(u) = \sigma(v)^2$. Applying the inverse shows squares map onto squares.

(b) Positive real numbers are exactly the nonzero squares. An automorphism preserves nonzero elements, so it maps positive reals to positive reals.

(c) If $a < b$, then $b - a > 0$, hence $\sigma(b) - \sigma(a) = \sigma(b-a) > 0$. Thus it preserves strict order.

(d) Every automorphism fixes $\mathbb{Q}$. If $\sigma(x) > x$, choose a rational q with $x < q < \sigma(x)$. Order preservation gives $\sigma(x) < \sigma(q) = q$, contradiction. If $\sigma(x) < x$, a rational between them gives the reverse contradiction. Hence $\sigma(x) = x$ for every real x.

49 The Isomorphism Extension Theorem

Exercise 49.1: Extending an identity on two radicals

An embedding of E is a sign choice on $\sqrt{2}, \sqrt{3}, \sqrt{5}$. Fixing $\sqrt{2}$ forces its sign to be positive; fixing $\sqrt{15} = \sqrt{3}\sqrt{5}$ forces the other two signs to agree. Thus the two triples of images are $(\sqrt{2}, \sqrt{3}, \sqrt{5})$ and $(\sqrt{2}, -\sqrt{3}, -\sqrt{5})$. Each is realized by a sign-change automorphism, and there are no others.

Exercise 49.2: Extending a prescribed negative product

Here $\sqrt{2}$ must stay fixed and the product $\sqrt{3}\sqrt{5}$ must change sign. Thus the two triples of images are $(\sqrt{2}, -\sqrt{3}, \sqrt{5})$ and $(\sqrt{2}, \sqrt{3}, -\sqrt{5})$. They both respect the prescribed map, and exhaust its possible extensions.

Exercise 49.3: Extending conjugation on a triple product

The product of the three radical signs must be negative. The four triples are

$$(-\sqrt{2}, \sqrt{3}, \sqrt{5}),\ (\sqrt{2}, -\sqrt{3}, \sqrt{5}),\ (\sqrt{2}, \sqrt{3}, -\sqrt{5}),\ (-\sqrt{2}, -\sqrt{3}, -\sqrt{5}).$$

Every triple defines a sign-change automorphism of E, and exactly the odd numbers of sign changes negate $\sqrt{30}$.

Exercise 49.4: Embeddings of a real cubic field

Put $r = \sqrt[3]{2}$ and $\omega = (-1 + i\sqrt{3})/2$. The polynomial $x^3 - 2$ is irreducible by Eisenstein and has distinct roots $r, r\omega, r\omega^2$. For each $j = 0, 1, 2$, there is one rational embedding sending r to $r\omega^j$. Its value on $a + br + cr^2$ is $a + br\omega^j + cr^2\omega^{2j}$. These are all three embeddings into $\overline{\mathbb{Q}}$.

Exercise 49.5: Adding an independent square root

Let r, ω be as in Exercise 49.4. The field $\mathbb{Q}(r)$ has degree 3 and cannot contain the quadratic field $\mathbb{Q}(\sqrt{3})$, by the tower law. Thus adjoining $\sqrt{3}$ doubles the degree. For each of the three choices $r \mapsto r\omega^j$, the extension theorem allows both roots $\sqrt{3} \mapsto \pm\sqrt{3}$ of its transported quadratic minimal polynomial. These six independent pairs describe all embeddings; a field map is determined by these generators.

Exercise 49.6: Extending a prescribed quadratic conjugation

The source field is $\mathbb{Q}(i, \sqrt{3}, r)$ and the map must send $\sqrt{3}$ to $-\sqrt{3}$. We can independently send i to $\pm i$ and r to $r\omega^j$ for $j = 0, 1, 2$. To justify existence, $\mathbb{Q}(i, \sqrt{3})$ is quadratic over $\mathbb{Q}(\sqrt{3})$, and $x^3 - 2$ remains irreducible over it: if a cubic root lay in this degree-four field, its degree-three rational subfield would violate the tower law. Thus there are exactly $2 \cdot 3 = 6$ extensions. For each, values on the three generators determine the entire map.

Exercise 49.7: A transcendental sign change

(a) The fixed field is $\mathbb{Q}(\pi^2)$. Indeed, $\mathbb{Q}(\pi)$ has basis $1, \pi$ over $\mathbb{Q}(\pi^2)$ because π is transcendental and $x^2 - \pi^2$ gives degree 2. The sign change fixes $A + B\pi$ precisely when $B = 0$.

(b) The element $\sqrt{\pi}$ satisfies $x^2 - \pi$ over $\mathbb{Q}(\pi)$. Transporting coefficients under $\pi \mapsto -\pi$ gives $x^2 + \pi$, whose roots in the algebraic closure are $\pm i\sqrt{\pi}$. The two extensions therefore send $\sqrt{\pi}$ to these two values. They exist by the extension theorem and are determined by this image.

Exercise 49.8: Ten statements about extensions of isomorphisms

(a) False. Take $F = \mathbb{Q}$ and α transcendental; no embedding of $\mathbb{Q}(\alpha)$ into the algebraic closure $\overline{\mathbb{Q}}$ can send a transcendental element to an algebraic one.

(b) True, by the isomorphism extension theorem for algebraic extensions.

(c) False. The identity on $\mathbb{Q}$ has one extension to $\mathbb{Q}$ but two to $\mathbb{Q}(\sqrt{2})$.

(d) True. An isomorphism extends to an isomorphism of algebraic closures.

(e) False. $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{2})$ are not isomorphic, but share an algebraic closure up to isomorphism.

(f) True. Both closures are algebraic closures of $\mathbb{Q}$.

(g) True. The number of embeddings is at most the finite degree.

(h) True. Count extensions of each lower embedding through a tower; each has the same number of extensions as the relative index.

(i) True. Successively extending across finitely many algebraic generators constructs the map; the infinite case needs the maximality argument.

(j) False. $\mathbb{C}$ is uncountable, whereas $\overline{\mathbb{Q}}$ is countable, so they cannot be isomorphic.

Exercise 49.9: An algebraic self-embedding of a closed field

The image $\sigma[K]$ is algebraically closed because it is isomorphic to the algebraically closed field K. By assumption every element of K is algebraic over this image. An algebraically closed field has no proper algebraic extension: each minimal polynomial is linear. Therefore every element of K already lies in $\sigma[K]$. Thus the embedding is onto and is an automorphism.

Exercise 49.10: Extending to an automorphism of the closure

Let $\sigma : E \to \overline{F}$ fix F. The extension theorem extends it to an embedding $\tau : \overline{F} \to \overline{F}$. Its image contains F, and $\overline{F}$ is algebraic over F, hence algebraic over the larger image $\tau[\overline{F}]$. Exercise 49.9 now makes τ onto. Its restriction to E is the desired original map.

Exercise 49.11: Closures of algebraically related fields

An algebraic closure $\overline{E}$ of the algebraic extension E/F is algebraic over F by transitivity of algebraicity. It is also algebraically closed. Thus it is an algebraic closure of F. Uniqueness of algebraic closures up to an isomorphism fixing F gives $\overline{E} \cong \overline{F}$.

Exercise 49.12: Closures of rational function fields

Since π is transcendental, $\sqrt{\pi}$ is transcendental too, and substitution $\sqrt{\pi} \mapsto x$ gives $\mathbb{Q}(\sqrt{\pi}) \cong \mathbb{Q}(x)$. Their algebraic closures are consequently isomorphic. The extension $\overline{\mathbb{Q}}(x)/\mathbb{Q}(x)$ is algebraic, so an algebraic closure of either is an algebraic closure of the other by Exercise 49.11. Finally the algebraic closure of $\mathbb{Q}(\sqrt{\pi})$ inside $\mathbb{C}$ is indeed algebraically closed: a root in $\mathbb{C}$ of a polynomial over it is algebraic over that field, hence algebraic over $\mathbb{Q}(\sqrt{\pi})$. These observations prove the requested isomorphism.

Exercise 49.13: Index is bounded by degree

For a simple algebraic extension $F(\alpha)/F$, the index is the number of distinct roots of the minimal polynomial, hence is at most its degree. Write a finite extension as a finite tower of simple extensions. Index and degree are both multiplicative along the tower, so multiplying the inequalities at each step gives $\{E : F\} \leq [E : F]$. Here braces denote the number of embeddings into a fixed algebraic closure, not a quotient field.

50 Splitting Fields

Exercise 50.1: Splitting a quadratic

The roots of x^2+3 are $\pm\sqrt{-3}$. Thus the splitting field is $\mathbb{Q}(\sqrt{-3})$ of degree 2, since the polynomial has no rational root.

Exercise 50.2: Fourth roots of unity

The roots of x^4-1 are $1,-1,i,-i$. Adjoining i suffices and is necessary. The splitting field $\mathbb{Q}(i)$ has degree 2 over $\mathbb{Q}$.

Exercise 50.3: Two distinct quadratic radicals

The splitting field is $\mathbb{Q}(\sqrt{2},\sqrt{3})$. It has degree 4: if $\sqrt{3}=a+b\sqrt{2}$ with rational a,b, squaring would give $3=a^2+2b^2+2ab\sqrt{2}$, hence $ab=0$. The alternatives would require either a rational square root of 3 or a rational square root of $3/2$, both impossible. Thus the second quadratic doubles the first degree.

Exercise 50.4: Splitting an irreducible cubic

Let $r=\sqrt[3]{3}$ and ω be a primitive cube root of unity. The splitting field is $\mathbb{Q}(r,\omega)$. Eisenstein at 3 gives $[\mathbb{Q}(r):\mathbb{Q}]=3$, and ω is nonreal while $\mathbb{Q}(r)\subset\mathbb{R}$. Its quadratic polynomial therefore doubles the degree, giving 6.

Exercise 50.5: Third roots of unity

Factor $x^3-1=(x-1)(x^2+x+1)$. The quadratic has discriminant -3, so its splitting field is $\mathbb{Q}(\sqrt{-3})=\mathbb{Q}(\omega)$, of degree 2.

Exercise 50.6: Combining square and cube roots

The roots require $\sqrt{2}, r=\sqrt[3]{2}$, and ω. Put $s=2^{1/6}$; then $s=\sqrt{2}/r$, while $s^2=r$ and $s^3=\sqrt{2}$. Thus the splitting field is $\mathbb{Q}(s,\omega)$. Eisenstein makes x^6-2 irreducible, so $[\mathbb{Q}(s):\mathbb{Q}]=6$. This real field does not contain the nonreal ω, giving total degree 12.

Exercise 50.7: Automorphisms of the real cubic field

The order is 1. An automorphism sends the real $r=\sqrt[3]{2}$ to a root of x^3-2 inside the real field $\mathbb{Q}(r)$. Its only real root is r, so every automorphism fixes this generator and is the identity.

Exercise 50.8: Automorphisms of its splitting field

The splitting field $E = \mathbb{Q}(r, i\sqrt{3})$ has degree 6 over $\mathbb{Q}$. Characteristic zero gives six embeddings, all of which are automorphisms because E is a splitting field. Thus the group order is 6.

Exercise 50.9: Fixing the real cubic generator

The extension $E/\mathbb{Q}(r)$ has degree 2, obtained by adjoining $i\sqrt{3}$. Its two automorphisms fix r and send $i\sqrt{3}$ to itself or its negative. Hence the relative group has order 2.

Exercise 50.10: A cubic splits in an eight-element field

The polynomial $f = x^3 + x^2 + 1$ has no root in $\mathbb{Z}_2$, so is irreducible. Thus $\mathbb{Z}_2(\alpha)$ has eight elements. Frobenius preserves roots of polynomials over $\mathbb{Z}_2$, so $\alpha, \alpha^2, \alpha^4$ are roots. They are distinct: a nonzero nonidentity element of the group of seven nonzero elements has order 7, and the exponents 1,2,4 are distinct modulo 7. Hence these exhaust its three roots and f splits there. From $\alpha^3 = \alpha^2 + 1$, the last root is $\alpha^4 = \alpha^2 + \alpha + 1$.

Exercise 50.11: Correcting the splitting-field criterion

In the stated condition, "every polynomial" must be replaced by "every irreducible polynomial." Otherwise even the splitting field $E = F = \mathbb{Q}$ fails the proposed condition for $(x - 1)(x^2 - 2)$. An algebraic extension E/F is a splitting field of a family of polynomials precisely when each irreducible polynomial over F with one zero in E has all its zeros in E.

Exercise 50.12: Correcting the meaning of splitting

A polynomial splits over E when it is a nonzero constant times a product of *linear* polynomials in $E[x]$. Merely factoring into lower-degree factors is insufficient: $(x^2 - 2)(x^2 - 3)$ factors over $\mathbb{Q}$ but does not split there.

Exercise 50.13: Bounds for a splitting-field degree

For a nonconstant degree-n polynomial, $1 \leq [E : F] \leq n!$. Adjoin a first root of degree at most n, factor out its linear factor, then adjoin a root of the remaining factor of degree at most $n - 1$, and continue. The tower law bounds the product by $n(n - 1) \cdots 1$. No irreducibility assumption gives a stronger positive lower bound: a polynomial already split over F has degree-one splitting field. For a nonzero constant polynomial the splitting field is F as well.

Exercise 50.14: Ten statements about splitting fields

(a) True. A conjugation isomorphism extends to the closure, and normality preserves E; the converse follows by applying an automorphism to a minimal polynomial.

(b) False. $\mathbb{R}/\mathbb{Q}$ is not algebraic, whereas a splitting field is algebraic.

(c) True: the trivial extension is the splitting field of x.

(d) True: $\mathbb{C}$ splits x^2+1 over $\mathbb{R}$ and is generated by its roots.

(e) True: it splits x^2+1 over $\mathbb{Q}$.

(f) True: $\mathbb{Q}(\pi)$ splits $x^2-\pi^2$ over $\mathbb{Q}(\pi^2)$.

(g) False. An isomorphism need not fix F or even have its image in the same field; send the transcendental generator of $E=F=\mathbb{Q}(t)$ to an independent indeterminate.

(h) False. For $E=F=\mathbb{Q}(t)$, the embedding $t\mapsto t^2$ has image $\mathbb{Q}(t^2)\subsetneq E$ and lies inside an algebraic closure of F.

(i) True. This is the defining embedding property of a normal algebraic extension.

(j) True. An algebraic closure is generated by all roots of all polynomials over F.

Exercise 50.15: Why irreducibility is essential

Take $F=E=\mathbb{Q}$, a splitting field of x over $\mathbb{Q}$. The reducible polynomial $(x-1)(x^2-2)$ has the zero 1 in E but its other two zeros are $\pm\sqrt{2}$, outside E. Thus the corollary about an irreducible polynomial having one root in a splitting field does not extend to reducible polynomials.

Exercise 50.16: Group orders in a tower

The intended multiplicativity formula has $|G(K/F)|$ on the left; the printed formula repeats $G(E/F)$ there.

(a) No. For $F=\mathbb{Q}$, $E=\mathbb{Q}(\sqrt[3]{2})$ and $K=\mathbb{Q}(\sqrt[3]{2},i\sqrt{3})$, the group orders are 6,1,2 respectively. Thus $6\neq 2\cdot 1$.

(b) Yes when the relevant fields are splitting fields over the bottom field. Their group orders equal their embedding indices, including for K/E, since a normal extension over F is normal over every intermediate field. The multiplicativity of indices then gives $|G(K/F)|=|G(K/E)||G(E/F)|$. This argument does not require separability.

Exercise 50.17: One polynomial suffices for a finite normal extension

Choose finitely many generators $\alpha_1,\ldots,\alpha_m$ for E/F and let p_j be their minimal polynomials over F. Since E is normal, each p_j splits in E. Let $f=\prod_j p_j$. Its splitting field inside E contains every α_j, so contains E; it is also contained in E, since all its roots are there. Hence it equals E.

Exercise 50.18: A quadratic extension is a splitting field

Choose $\alpha\in E\setminus F$. The tower law gives $E=F(\alpha)$ and a degree-two minimal polynomial x^2+bx+c. Its other root, counted with multiplicity, is $-b-\alpha\in E$. Thus the polynomial splits in E, and one root generates E, proving the claim even in characteristic 2 (where the roots may coincide).

Exercise 50.19: Containing all conjugates

If E/F is a splitting field, every irreducible polynomial with a root in E splits there, so all conjugates of every element belong to E. Conversely, for each $\alpha \in E$, take its minimal polynomial. The hypothesis places all its roots in E. The splitting field of this family is contained in E and contains every α, hence is exactly E.

Exercise 50.20: No nontrivial automorphism of the real cubic

Every automorphism of $\mathbb{Q}(r)$ fixes $\mathbb{Q}$, and the image of $r = \sqrt[3]{2}$ must satisfy $x^3 - 2 = 0$. The other two roots $r\omega, r\omega^2$ are nonreal, whereas the field is real. Thus r is fixed. Since it generates the field, the automorphism is the identity.

Exercise 50.21: The cyclic subgroup fixing the quadratic radical

Put $r = \sqrt[3]{2}$ and $\omega = (-1 + i\sqrt{3})/2$. The field $\mathbb{Q}(i\sqrt{3}) = \mathbb{Q}(\omega)$ has degree 2 over $\mathbb{Q}$, so its relative degree in the degree-six splitting field is 3. The map $\rho(r) = r\omega$, $\rho(\omega) = \omega$ is an automorphism, and its powers send r successively to $r\omega^2$ and r. It has order 3 and accounts for all three relative embeddings. Thus the group is cyclic, isomorphic to $(\mathbb{Z}_3, +)$.

Exercise 50.22: The permutation representation on roots

(a) For an automorphism *fixing* F, $f(\sigma(\alpha)) = \sigma(f(\alpha)) = 0$. Its inverse shows that this gives a permutation of the roots. The base-field qualification is necessary.

(b) The roots generate the splitting field over F. Agreement on all roots and on F implies agreement on every rational expression in them, hence on the whole field.

(c) Restriction to the roots is therefore an injective homomorphism $G(E/F) \hookrightarrow S_\Omega$, where Ω is the set of distinct roots. Its image is the subgroup of permutations that preserve all algebraic relations among the roots over F.

Exercise 50.23: The full cubic permutation group

(a) The field $E = \mathbb{Q}(r, \omega)$ has degree 6 and is separable in characteristic zero. All six rational embeddings are automorphisms, so $|G(E/\mathbb{Q})| = 6$.

(b) Its action on the three roots gives an injection into S_3. As both groups have six elements, the injection is onto, proving $G(E/\mathbb{Q}) \cong S_3$.

Exercise 50.24: A prime-root-of-unity extension

For a prime p, every nonidentity pth root ζ is primitive, and its powers give all roots of $x^p - 1$. Thus its splitting field is $\mathbb{Q}(\zeta)$. The minimal polynomial $1 + x + \cdots + x^{p-1}$ is irreducible over $\mathbb{Q}$ by the prime cyclotomic theorem, so the extension degree is $p - 1$ (including degree 1 for $p = 2$).

Exercise 50.25: Uniqueness of splitting fields in different closures

There is an isomorphism $\tau : \overline{F} \to \overline{F}'$ fixing F. It maps roots of f to roots of f because the coefficients are fixed. Its inverse gives the reverse inclusion, so it bijects the two full root sets. Since each splitting field is generated by the respective roots, $\tau(E) = E'$. The restriction is the desired F-isomorphism.

51 Separable Extensions

Exercise 51.1: Combining square and cube roots

Choose $\alpha = \sqrt{2}/\sqrt[3]{2} = 2^{1/6}$. This quotient is in the given field. Direct exponent calculations give $\alpha^3 = \sqrt{2}$ and $\alpha^2 = \sqrt[3]{2}$, which are polynomials in α with rational coefficients. Thus each original generator lies in $\mathbb{Q}(\alpha)$, while α lies in their field, proving equality.

Exercise 51.2: Combining fourth and sixth roots

Choose $\alpha = \sqrt[4]{2}/\sqrt[6]{2} = 2^{1/12}$, an element of the given field. Then $\sqrt[4]{2} = \alpha^3$ and $\sqrt[6]{2} = \alpha^2$. These explicit polynomial expressions prove $\mathbb{Q}(\sqrt[4]{2}, \sqrt[6]{2}) = \mathbb{Q}(\alpha)$.

Exercise 51.3: A primitive biquadratic generator

Let $\alpha = \sqrt{2} + \sqrt{3}$, which belongs to the given field. Cubing and simplifying gives $\alpha^3 = 11\sqrt{2} + 9\sqrt{3}$. Subtracting suitable multiples of α yields

$$\sqrt{2} = \frac{\alpha^3 - 9\alpha}{2}, \qquad \sqrt{3} = \frac{11\alpha - \alpha^3}{2}.$$

Thus both generators are rational polynomials in α, giving the desired equality.

Exercise 51.4: A primitive generator for a complex sextic field

Put $r = \sqrt[3]{2}$ and choose $\alpha = r + i$. From $(\alpha - i)^3 = 2$ and $i^2 = -1$,

$$i = \frac{\alpha^3 - 3\alpha - 2}{3\alpha^2 - 1}.$$

The denominator is nonzero: α has imaginary part 1 and positive real part, so its square has nonzero imaginary part. This already proves that $i, r \in \mathbb{Q}(\alpha)$. To meet the requested *polynomial* form, exact reduction gives

$$i = \frac{-12\alpha^5 + 9\alpha^4 - 40\alpha^3 + 78\alpha^2 - 78\alpha - 91}{22},$$
$$r = \frac{12\alpha^5 - 9\alpha^4 + 40\alpha^3 - 78\alpha^2 + 100\alpha + 91}{22}.$$

Here is a way to check these expressions without assuming the answer. Squaring the first displayed numerator equation for i gives

$$m(\alpha) = 0, \quad m(x) = x^6 + 3x^4 - 4x^3 + 3x^2 + 12x + 5.$$

Polynomial division verifies that $(3x^2 - 1)$ times the displayed polynomial for i is congruent to $x^3 - 3x - 2$ modulo $m(x)$. Evaluation at α proves the formula; subtracting it from α gives the formula for r. Hence the field is $\mathbb{Q}(\alpha)$.

Exercise 51.5: Where multiplicity is measured

The factorization must be in $\overline{F}[x]$, not necessarily in $F[x]$, since α may lie outside F. For a *nonzero* polynomial f, multiplicity $v \geq 1$ means $(x-\alpha)^v \mid f$ but $(x-\alpha)^{v+1} \nmid f$ in $\overline{F}[x]$. The zero polynomial has no finite root multiplicity under this definition.

Exercise 51.6: The definition of separable element

The proposed definition is correct: an algebraic element is separable over F when it has multiplicity 1 as a root of its minimal polynomial over F. All roots of an irreducible polynomial have the same multiplicity, so this also says the minimal polynomial has distinct roots in an algebraic closure.

Exercise 51.7: A repeated polynomial over a perfect field

Take $f(x) = (x^2+1)^2$. It has no rational root, and its complex roots $i, -i$ each have multiplicity 2. This does not contradict perfection of $\mathbb{Q}$, which concerns *irreducible* polynomials. The minimal polynomial x^2+1 has distinct roots; the repetitions were introduced by squaring it.

Exercise 51.8: Ten statements about separability

(a) False. Over $\mathbb{Z}_p(t)$, $x^p - t$ is irreducible and has just one distinct root, so adjoining that root gives an inseparable finite extension.

(b) True. Every finite field is perfect.

(c) True. An irreducible polynomial in characteristic zero cannot share a factor with its smaller-degree nonzero derivative.

(d) False. $(x-1)^2$ has a repeated root over any field.

(e) False for the same example, even over $\mathbb{Q}$.

(f) True. This is the defining polynomial characterization of a perfect field.

(g) True. Every irreducible polynomial over an algebraically closed field is linear.

(h) True. An algebraic closure is algebraic over F and is perfect.

(i) True. Separability makes the embedding index equal the degree, and splitting makes every embedding an automorphism.

(j) True. The index of a finite extension divides its degree (the quotient is its inseparable degree), and in a splitting field the index equals the group order.

Exercise 51.9: Separable elements are closed under field operations

The minimal polynomial of β over $F(\alpha)$ divides its separable minimal polynomial over F, so it is separable. Therefore $F(\alpha,\beta)/F(\alpha)$ and $F(\alpha)/F$ are finite separable extensions; the tower theorem makes $F(\alpha,\beta)/F$ separable. Every element of this field is then separable over F, including $\alpha+\beta$, $\alpha\beta$, and α/β when $\beta \neq 0$.

Exercise 51.10: A basis for a Frobenius extension

Let $K = \mathbb{Z}_p(y^p)$. The relation $y^p \in K$ makes $K[y]$ spanned by $1, y, \ldots, y^{p-1}$. Being a finite-dimensional domain over K, it is a field: multiplication by any nonzero element is an injective linear map, hence surjective, giving an inverse. Thus $K[y] = \mathbb{Z}_p(y)$. For independence, a relation $\sum_{j=0}^{p-1} r_j(y^p)y^j = 0$ can be multiplied by a common denominator to give $\sum_j P_j(y^p)y^j = 0$. Exponents coming from different j lie in different residue classes modulo p, so no cancellation between the groups is possible; all P_j vanish. Thus the displayed spanning set is a basis, and the degree is p. Setting $t = y^p$, the polynomial $x^p - t$ has y as a root and the same degree as its minimal polynomial. Hence it is irreducible over $\mathbb{Z}_p(t)$.

Exercise 51.11: Algebraic extensions of a perfect field

Let α be algebraic over E. Since E/F is algebraic, α is algebraic over F as well. The minimal polynomial of α over the perfect field F is separable. Its minimal polynomial over E divides it in $E[x]$, so also has no repeated roots. Thus every element algebraic over E is separable over E, proving E perfect.

Exercise 51.12: Transitivity for infinite separable extensions

Take $\alpha \in K$. Its separable minimal polynomial over E uses finitely many coefficients $c_1, \ldots, c_m \in E$. Set $E_0 = F(c_1, \ldots, c_m)$; this is a finite separable extension of F. The same polynomial is irreducible over E_0, because a factorization there would be one over E, and it remains separable. Thus $E_0(\alpha)/E_0$ is finite separable. The finite tower theorem gives $E_0(\alpha)/F$ separable, so α is separable over F. This holds for every $\alpha \in K$, proving transitivity without assuming the entire tower finite.

Exercise 51.13: The separable closure inside an algebraic extension

Let S be the elements of E separable over F. It contains F, including 0 and 1. Exercise 51.9 shows it is closed under sums, products, and quotients by nonzero elements; additive inverses are included by multiplying by $-1 \in F$. Hence S is a subfield. It consists exactly of the separable elements, so is the largest intermediate extension separable over F, called the separable closure of F in E.

Exercise 51.14: Frobenius generates a finite-field Galois group

(a) In a field of p^n elements, $a^{p^n} = a$ for every a, so $\sigma_p^n = 1$. If $\sigma_p^d = 1$ for $0 < d < n$, all p^n elements would be roots of $x^{p^d} - x$, whose degree p^d is smaller than p^n, impossible. Thus its order is n.

(b) The extension has degree n, is separable because finite fields are perfect, and is a splitting extension since it splits $x^{p^n} - x$. Its Galois group therefore has order n. The n distinct powers of Frobenius already account for that many elements, so the full group is cyclic with this generator.

Exercise 51.15: The additive derivative homomorphism

(a) If $f = \sum a_j x^j$ and $g = \sum b_j x^j$, differentiating coefficient by coefficient gives $(f+g)' = \sum j(a_j + b_j)x^{j-1} = f' + g'$. This proves it is a homomorphism of additive groups; it is generally not a

ring homomorphism.

(b) In characteristic zero, $ja_j = 0$ for $j > 0$ implies $a_j = 0$. Thus the kernel is the constant polynomials F.

(c) In characteristic p, j vanishes as a field scalar precisely when $p \mid j$. Hence the kernel is $F[x^p]$, the polynomials all of whose nonzero exponents are multiples of p. Their coefficients need not themselves be pth powers.

Exercise 51.16: Formal derivative rules

(a) For $c \in F$, coefficientwise differentiation gives $(cf)' = \sum_j jca_j x^{j-1} = cf'$.

(b) First check $(x^i x^j)' = (i+j)x^{i+j-1} = x^i(jx^{j-1}) + (ix^{i-1})x^j$. Expand arbitrary polynomials into monomials and use additivity and (a) to obtain $(fg)' = fg' + f'g$.

(c) Induct on m. For $m = 1$ the assertion is immediate. If $(f^m)' = mf^{m-1}f'$, the product rule gives $(f^{m+1})' = f^m f' + (f^m)'f = (m+1)f^m f'$. The integers are interpreted as elements of F, so this proof also works in positive characteristic.

Exercise 51.17: Multiplicity and vanishing derivative

In $\overline{F}[x]$, write $f = (x-\alpha)^v g$ with $g(\alpha) \neq 0$. If $v = 1$, the product rule gives $f'(\alpha) = g(\alpha) \neq 0$. If $v > 1$, both terms of

$$f' = v(x-\alpha)^{v-1}g + (x-\alpha)^v g'$$

vanish at α, including when the scalar v is zero in the field. Thus $f'(\alpha) = 0$ exactly when the root has multiplicity greater than 1.

Exercise 51.18: Characteristic zero forces irreducible separability

Let $p \in F[x]$ be nonconstant and irreducible in characteristic zero. Its derivative is nonzero and has smaller degree. If a root α were repeated, it would be a root of p' too, so its minimal polynomial p would divide p'. The degree inequality rules this out. Therefore all roots of p are simple, proving separability.

Exercise 51.19: The derivative test for an inseparable irreducible

For an irreducible q, a repeated root implies $q \mid q'$ by minimality. Since $\deg q' < \deg q$ unless $q' = 0$, this is equivalent to $q' = 0$. In characteristic p, coefficientwise differentiation gives $q' = 0$ exactly when each exponent with nonzero coefficient is divisible by p. Conversely a nonconstant polynomial with zero derivative cannot have a simple root by Exercise 51.17. Thus the criterion holds in both directions.

Exercise 51.20: Common factors over the algebraic closure

By Exercise 51.17 a repeated root is exactly a common root of f and f'. Over the algebraically closed field $\overline{F}$, a common root α supplies the common linear factor $x - \alpha$. Conversely a nonconstant

common factor has a root in the closure, which is a common root of f, f', hence repeated in f. This proves the requested equivalence for nonzero f.

Exercise 51.21: Bringing the gcd test back to the coefficient field

If f, f' have no nonconstant common divisor in $F[x]$, their gcd is 1, so Bezout's identity gives $Af + Bf' = 1$ with $A, B \in F[x]$. This same identity holds in $\overline{F}[x]$, excluding any nonconstant common factor there. Conversely a nonconstant common factor in $F[x]$ remains one over the closure. Combining with Exercise 51.20 proves the equivalence over F itself. The argument includes nonzero constant f, whose gcd with its zero derivative is a unit.

Exercise 51.22: A root-free computational test

For a nonzero polynomial, form its formal derivative and run the Euclidean algorithm in $F[x]$ on f, f'. Normalize the final nonzero remainder to a monic gcd d. A repeated root exists exactly when $\deg d > 0$. In particular, if f is nonconstant and $f' = 0$, the gcd is the monic associate of f, so there is repetition. If f is a nonzero constant, the gcd has degree 0 and there are no roots. No numerical root approximation is required.

52 Totally Inseparable Extensions

For degree checks involving two independent indeterminates, one useful elementary rule is

$$[\mathbb{F}_3(y,z) : \mathbb{F}_3(y^a z^b, y^c z^d)] = |ad - bc|$$

when the two exponent vectors are independent. Here is why the determinant appears. The subgroup of $\mathbb{Z}^2$ generated by $(a,b),(c,d)$ has $|ad-bc|$ cosets. Choose one monomial for each exponent coset. Every Laurent monomial is a coefficient-field monomial times one chosen representative, so these representatives span; their span is a finite-dimensional domain over the coefficient field, hence is a field and includes rational functions too. They are independent because clearing coefficient denominators leaves terms in distinct exponent cosets, which cannot cancel. Thus they form a basis. This supplies the degree certificates below; separability and the power tests establish the actual closure fields.

Exercise 52.1: A separable closure with two independent exponents

Let $K = \mathbb{Z}_3(y^{12}, z^{18})$ and $L = \mathbb{Z}_3(y,z)$. Set $S = \mathbb{Z}_3(y^3, z^9)$. The generators y^3 and z^9 satisfy respectively $X^4 - y^{12}$ and $X^2 - z^{18}$ over K, whose nonzero roots are simple in characteristic 3. Thus S/K is separable. Also $y^3, z^9 \in S$, so every element of L has some 3-power in S; in fact its ninth power is in S. Hence L/S is purely inseparable. An element separable over K is separable over S too, since its new minimal polynomial divides its old separable one; if it also lies in the purely inseparable extension L/S, it must be in S. Therefore S is exactly the separable closure. The monomial degree counts are $[L:K] = 12 \cdot 18 = 216$, $[L:S] = 3 \cdot 9 = 27$, and $[S:K] = 8$.

Exercise 52.2: A separable closure with a mixed monomial

Now $K = \mathbb{Z}_3(y^{12}, y^2 z^{18})$. Put $S = \mathbb{Z}_3(y^3, yz^9)$. The first generator satisfies the separable equation $X^4 - y^{12} = 0$ and the second the separable equation $X^2 - y^2 z^{18} = 0$, so S/K is separable. Moreover $y^3 \in S$ and $z^{27} = (yz^9)^3/y^3 \in S$. Thus $L = \mathbb{Z}_3(y,z)$ is purely inseparable over S. The separable/purely-inseparable intersection argument from Exercise 52.1 makes S the full separable closure. For a degree check, the exponent lattice of K has determinant $12 \cdot 18 = 216$, while that of S, generated by $(3,0),(1,9)$, has determinant 27; hence $[S:K] = 8$.

Exercise 52.3: A totally inseparable closure in the independent case

The field is $P = \mathbb{Z}_3(y^4, z^2)$. Its generators have a third and a ninth power, respectively, in $K = \mathbb{Z}_3(y^{12}, z^{18})$, so P/K is purely inseparable. The extension L/P is separable, because y satisfies $X^4 - y^4$ and z satisfies $X^2 - z^2$, both separable polynomials. If $a \in L$ has a 3-power in K, it is purely inseparable over P but also separable over P, so $a \in P$. This proves maximality. Thus the totally inseparable closure is P; its degree over K is 27.

Exercise 52.4: A totally inseparable closure in the mixed case

Take $P = \mathbb{Z}_3(y^4, y^2 z^2)$. It contains $K = \mathbb{Z}_3(y^{12}, y^2 z^{18})$, since $y^{12} = (y^4)^3$ and $y^2 z^{18} = (y^2 z^2)^9/(y^4)^4$. Its generators have 3-powers in K:

$$(y^4)^3 = y^{12}, \qquad (y^2 z^2)^{27} = (y^{12})^4 (y^2 z^{18})^3.$$

Hence P/K is purely inseparable. Over P, first adjoining y is separable via its fourth-power equation, and then adjoining z is separable via $z^2 = (y^2z^2)/y^2$. Thus L/P is separable. The same intersection argument as in Exercise 52.3 shows that every element of L purely inseparable over K lies in P. Its exponent lattice, generated by $(4,0),(2,2)$, has determinant 8, so $[P:K] = 216/8 = 27$.

Exercise 52.5: Ten statements about totally inseparable extensions

In this section Fraleigh reserves "totally inseparable" for a *proper* extension with one embedding. The more common "purely inseparable" convention also permits the trivial extension.

(a) False. $\mathbb{Z}_3(t)(\sqrt{t})/\mathbb{Z}_3(t)$ is a proper separable extension, since $X^2 - t$ is irreducible and has nonzero derivative.

(b) True. A totally inseparable algebraic element has some p^t-power in F.

(c) False. $\mathbb{Z}_5(y)/\mathbb{Z}_5(y^5)$ has degree 5 and only one embedding; its minimal polynomial is $X^5 - y^5$.

(d) False. $\mathbb{Z}_5(y)/\mathbb{Z}_5(y^{10})$ has inseparable degree 5 and separable degree 2, since $X^{10} - y^{10}$ has two distinct roots, each repeated five times.

(e) False. The same mixed extension has two embeddings, so is not totally inseparable either.

(f) False. The element y over $\mathbb{Z}_5(y^{10})$ in (d) is neither separable nor totally inseparable.

(g) True. The separable elements form a subfield by Exercise 51.13.

(h) False with the book's strict convention: if E/F is separable, its separable closure is E, and E/E is not a proper extension. The corrected claim allows E to equal that closure, or uses the inclusive "purely inseparable" convention.

(i) True. If E/F is not separable, its separable closure is proper, and every element outside it is totally inseparable over it.

(j) True, by the minimal-polynomial characterization of total inseparability.

Exercise 52.6: Purely inseparable elements form a field

Include the elements of F as requested. For a, b in the proposed set, choose a common power $q = p^m$ with $a^q, b^q \in F$. Then $(a+b)^q = a^q + b^q \in F$, $(ab)^q = a^q b^q \in F$, and if $b \neq 0$, $(a/b)^q = a^q/b^q \in F$. Additive inverses work in the same way. Thus it is closed under field operations and contains 0,1. Every element outside F with such a power is totally inseparable, by the power criterion, so these closure calculations prove the assertion. In characteristic zero no proper totally inseparable elements exist, and the set is simply F.

Exercise 52.7: Perfection and surjective Frobenius

If F is perfect and $a \in F$, choose a root b of $X^p - a$ in a closure. That polynomial equals $(X-b)^p$, so b has only one conjugate. Separability forces its degree to be 1; hence $b \in F$ and $a = b^p$. Thus $F^p = F$. Conversely suppose $F^p = F$. An inseparable irreducible polynomial would have derivative zero, so have form $\sum a_j X^{pj}$. Write $a_j = b_j^p$. Then it equals $(\sum b_j X^j)^p$, contradicting irreducibility. Therefore all irreducibles are separable, and F is perfect.

Exercise 52.8: Perfection through a finite extension

Let $n = [E : F]$. Frobenius is an isomorphism from E onto E^p carrying F onto F^p, so $[E^p : F^p] = n$. If $F^p = F$, then $F \subseteq E^p \subseteq E$ and both $[E^p : F]$ and $[E : F]$ equal n, forcing $E^p = E$. Conversely, if $E^p = E$, the equality above reads $[E : F^p] = n$. In the tower $F^p \subseteq F \subseteq E$, it follows that $n = n[F : F^p]$, so $[F : F^p] = 1$ and $F^p = F$. Finiteness of n is essential for this degree argument.

53 Galois Theory

Exercise 53.1: The embedding index of a multiquadratic field

The characteristic is zero, so the extension is separable. Thus its index, the number of embeddings fixing $\mathbb{Q}$, equals its given degree: $\{K : \mathbb{Q}\} = 8$.

Exercise 53.2: The group order of a multiquadratic field

All independent sign changes of $\sqrt{2}, \sqrt{3}, \sqrt{5}$ are automorphisms. They give $2^3 = 8$ maps, exhausting the degree bound. Thus $|G(K/\mathbb{Q})| = 8$.

Exercise 53.3: The subgroup fixing the bottom field

By definition $\lambda(\mathbb{Q}) = G(K/\mathbb{Q})$, so its order is 8. The whole rational Galois group fixes the bottom field.

Exercise 53.4: Fixing two specified radicals

The field $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ has degree 4 over $\mathbb{Q}$. The subgroup fixing it has order $[K : \mathbb{Q}(\sqrt{2}, \sqrt{3})] = 8/4 = 2$, consisting of the identity and the sign change of $\sqrt{5}$.

Exercise 53.5: Fixing a product radical

The field $\mathbb{Q}(\sqrt{6})$ has degree 2 over $\mathbb{Q}$, so its fixing subgroup has order $8/2 = 4$. Equivalently the signs of $\sqrt{2}$ and $\sqrt{3}$ must agree, while the sign of $\sqrt{5}$ is arbitrary.

Exercise 53.6: Fixing a triple-product radical

The quadratic field $\mathbb{Q}(\sqrt{30})$ has degree 2 over $\mathbb{Q}$, so its fixing subgroup has order 4. Its sign changes are exactly the choices with a positive product of the three radical signs.

Exercise 53.7: Fixing a sum of two radicals

Put $\alpha = \sqrt{2} + \sqrt{6}$. Its square is $8 + 4\sqrt{3}$, so $\sqrt{3} = (\alpha^2 - 8)/4 \in \mathbb{Q}(\alpha)$. Also $\sqrt{2} = \alpha/(1 + \sqrt{3})$ lies in this field. Thus $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ has degree 4; its fixing subgroup has order $8/4 = 2$.

Exercise 53.8: Fixing the entire top field

An automorphism fixing every element of K is the identity. Therefore $\lambda(K)$ is the trivial subgroup, of order 1.

Exercise 53.9: The group of fourth roots of unity

The splitting field of $x^4 - 1$ is $\mathbb{Q}(i)$. The two automorphisms send i to i and $-i$. Their composition table is that of the cyclic group $\mathbb{Z}_2$; the nonidentity map is complex conjugation.

Exercise 53.10: A relative finite-field Frobenius

Since $729 = 9^3$, the degree of GF(729) over GF(9) is 3. Its Galois group is cyclic of order 3, generated by $a \mapsto a^9$. The third power is $a \mapsto a^{729} = a$; a smaller positive power cannot fix all 729 elements by the polynomial root bound.

Exercise 53.11: The cubic splitting-field correspondence

Let $r = \sqrt[3]{2}$, $\omega = (-1 + i\sqrt{3})/2$, $K = \mathbb{Q}(r, \omega)$, $\rho(r) = r\omega$, $\rho(\omega) = \omega$, and let κ be complex conjugation.

(a) The six maps and their images are

σ	$\sigma(r)$	$\sigma(i\sqrt{3})$
1	r	$i\sqrt{3}$
ρ	$r\omega$	$i\sqrt{3}$
ρ^2	$r\omega^2$	$i\sqrt{3}$
κ	r	$-i\sqrt{3}$
$\rho\kappa$	$r\omega$	$-i\sqrt{3}$
$\rho^2\kappa$	$r\omega^2$	$-i\sqrt{3}$

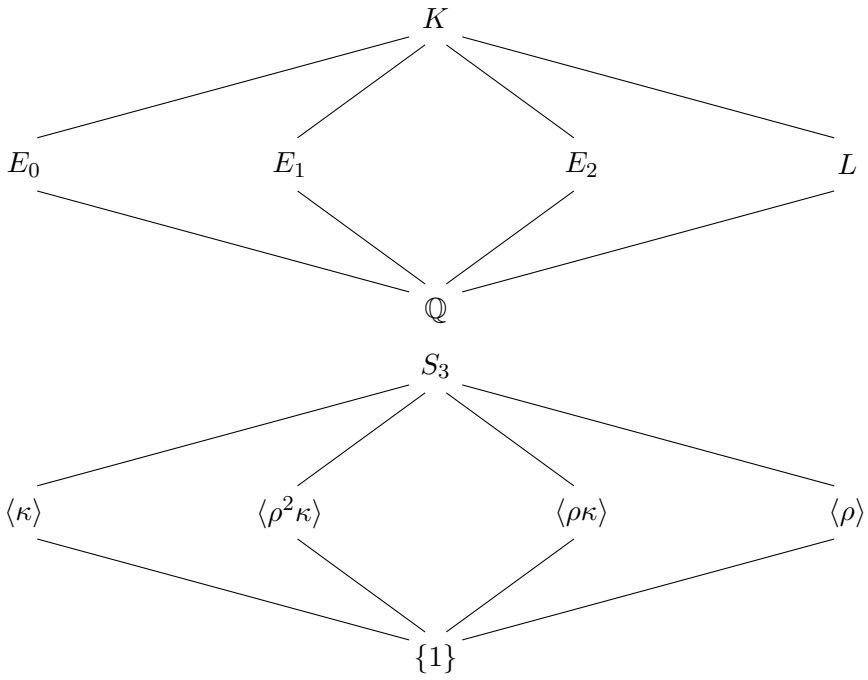
These account for all six rational embeddings of the degree-six splitting field.

(b) Their faithful action on the three cubic roots identifies the group with S_3. In particular $\rho^3 = \kappa^2 = 1$ and $\kappa\rho\kappa = \rho^{-1}$.

(c) The correspondence is

subgroup	fixed field
S_3	$\mathbb{Q}$
$\langle\rho\rangle$	$L = \mathbb{Q}(\omega)$
$\langle\kappa\rangle$	$E_0 = \mathbb{Q}(r)$
$\langle\rho\kappa\rangle$	$E_2 = \mathbb{Q}(r\omega^2)$
$\langle\rho^2\kappa\rangle$	$E_1 = \mathbb{Q}(r\omega)$
$\{1\}$	K

For instance $\rho\kappa(r\omega^2) = r\omega \cdot \omega^{-2} = r\omega^2$. The field degrees (3 for the E_j, 2 for L) and the Galois correspondence prove these are the full fixed fields. The diagrams below have only the displayed covering relations; invert one to obtain the other.



Exercise 53.12: The group of two quadratic factors

Since $x^4 - 5x^2 + 6 = (x^2 - 2)(x^2 - 3)$, its splitting field is $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ of degree 4. The independent sign flips of the two radicals commute and each has order 2. Thus its group is $\mathbb{Z}_2 \times \mathbb{Z}_2$, the Klein four-group.

Exercise 53.13: The group of third roots of unity

The splitting field is $\mathbb{Q}(\omega)$ with $\omega^2 + \omega + 1 = 0$. Its two maps send ω to ω and ω^2, so the group is cyclic of order 2.

Exercise 53.14: An abstract group does not determine its field

Take $K_1 = \mathbb{Q}(\sqrt{2})$ and $K_2 = \mathbb{Q}(\sqrt{3})$ over $F = \mathbb{Q}$. Both have Galois group $\mathbb{Z}_2$. They are not isomorphic as fields: any isomorphism fixes $\mathbb{Q}$ and would send $\sqrt{2}$ to a square root of 2 in K_2. Writing $(a + b\sqrt{3})^2 = 2$ forces $ab = 0$, leading to $a^2 = 2$ or $b^2 = 2/3$, neither with rational solutions. Thus such an image does not exist.

Exercise 53.15: Ten statements about the Galois correspondence

Here "finite normal" has Fraleigh's definition: a *separable splitting extension*, often called a finite Galois extension.

(a) False. The subgroup-to-fixed-field correspondence is one-to-one.

(b) False. Inclusion reverses: $E \subseteq L$ implies $\lambda(L) \subseteq \lambda(E)$.

(c) True. The same splitting polynomials work over E, and separability descends along the tower.

(d) True. Each degree equals the order of its Galois group.

(e) True. A normal subgroup corresponds exactly to a normal intermediate extension of F.

(f) False. $\mathbb{Q}(\zeta_5)/\mathbb{Q}$ is simple but has cyclic group of order 4, which has a proper subgroup of order 2.

(g) False. A quadratic separable extension has simple Galois group $\mathbb{Z}_2$.

(h) True. Finite-field Galois groups are cyclic.

(i) False in characteristic 2 with this definition. $\mathbb{Z}_2(t^{1/2})/\mathbb{Z}_2(t)$ has degree 2 but is inseparable, hence is not "finite normal" as Fraleigh defines it (it is normal in the modern splitting-only sense).

(j) True. A quadratic in characteristic different from 2 is separable and splits in its root field.

Exercise 53.16: Abelian Galois extensions in a tower

The group $G(K/E)$ is a subgroup of the abelian group $G(K/F)$, hence is abelian. If E/F is normal, restriction gives $G(E/F) \cong G(K/F)/G(K/E)$. A quotient of an abelian group is abelian. Thus K/E and E/F are both abelian extensions in the requested sense. In fact every subgroup of an abelian group is normal, so the normality of intermediate E/F follows automatically here.

Exercise 53.17: Norm and trace lie in the base field

For any $\tau \in G(K/F)$, left multiplication $\sigma \mapsto \tau\sigma$ permutes the group. Hence

$$\tau\left(\prod_{\sigma}\sigma(\alpha)\right) = \prod_{\sigma}(\tau\sigma)(\alpha) = \prod_{\sigma}\sigma(\alpha),$$

and the analogous equality holds for the sum. Thus norm and trace are fixed by the entire group. Its fixed field is exactly F by the Galois correspondence, so both belong to F.

Exercise 53.18: Eight norm and trace calculations

The four embeddings independently change the signs of $\sqrt{2}, \sqrt{3}$.

(a) $N(\sqrt{2}) = (\sqrt{2})^2(-\sqrt{2})^2 = 4$.

(b) $N(\sqrt{2}+\sqrt{3}) = (\sqrt{2}+\sqrt{3})(\sqrt{2}-\sqrt{3})(-\sqrt{2}+\sqrt{3})(-\sqrt{2}-\sqrt{3}) = 1$, since the first pair is -1 and the second pair is -1.

(c) $N(\sqrt{6}) = (\sqrt{6})^2(-\sqrt{6})^2 = 36$.

(d) Every embedding fixes 2, so $N(2) = 2^4 = 16$.

(e) The two copies each of $\sqrt{2}, -\sqrt{2}$ sum to 0.

(f) The four sign choices of $\sqrt{2}+\sqrt{3}$ sum to 0.

(g) The two copies each of $\sqrt{6}, -\sqrt{6}$ sum to 0.

(h) The four copies of 2 sum to 8.

Exercise 53.19: Norm and trace from minimal-polynomial coefficients

Because $K = F(\alpha)$ is finite normal, its n automorphisms give exactly the n distinct roots of the minimal polynomial of α. Thus

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = \prod_{\sigma \in G(K/F)} (x - \sigma(\alpha)).$$

Comparing constant terms gives $a_0 = (-1)^n \prod_\sigma \sigma(\alpha)$, so (a) $N_{K/F}(\alpha) = (-1)^n a_0$. Comparing the coefficient of x^{n-1} gives (b) $T_{K/F}(\alpha) = -a_{n-1}$.

Exercise 53.20: The group order divides the factorial

Let Ω be the set of distinct roots of f in its splitting field K, and let $m = |\Omega| \leq n$. Because all irreducible factors are separable, K/F is finite normal in the book's sense. The faithful action of $G(K/F)$ on its generating roots embeds it into S_m. Lagrange's theorem gives $|G(K/F)| \mid m!$, and $m! \mid n!$. Hence the order divides $n!$. Repeated irreducible factors in f cause no difficulty when one uses distinct roots.

Exercise 53.21: The natural permutation group of a polynomial

For $\sigma \in G(K/F)$, its values on the roots define a permutation, since applying σ to $f(\alpha) = 0$ gives $f(\sigma\alpha) = 0$. Restriction respects composition. Its kernel fixes every root, hence fixes the field they generate over F, namely K, so its kernel is trivial. Thus the Galois group is naturally isomorphic to its image, a subgroup of the permutations of the distinct roots.

Exercise 53.22: Cyclotomic extensions over a general field

(a) The powers of a primitive nth root ζ give all roots of $x^n - 1$. Its derivative nx^{n-1} has no common root with it under the characteristic hypothesis. Thus $F(\zeta)$ is a finite separable splitting extension, hence finite normal.

(b) Any automorphism sends ζ to another primitive root ζ^a, where a is invertible modulo n. The map assigning a to that automorphism is injective, and composition multiplies exponents modulo n. It embeds the group into the abelian unit group $(\mathbb{Z}/n\mathbb{Z})^\times$. Therefore the Galois group is abelian, even when not every invertible exponent occurs.

Exercise 53.23: Cyclic extensions and their intermediate fields

(a) A subgroup of a cyclic group is cyclic, so $G(K/E)$ is cyclic. If E/F is normal, its group is the quotient $G(K/F)/G(K/E)$, also cyclic. In fact the normality follows automatically because every subgroup of a cyclic group is normal.

(b) If $|G(K/F)| = n$, a cyclic group has exactly one subgroup of each order dividing n. An intermediate field of degree d over F corresponds to a subgroup of index d, or order n/d. Hence for each divisor d of n there is exactly one such intermediate field.

Exercise 53.24: The polynomial formed from every automorphism

(a) Each $\tau \in G(K/F)$ permutes the factors of $f(x) = \prod_\sigma (x - \sigma\alpha)$, since $\sigma \mapsto \tau\sigma$ permutes the group. Thus every coefficient is fixed by the group, and belongs to F.

(b) All distinct conjugates of α occur, by the extension theorem, and each occurs the same number of times: the maps taking α to a specified conjugate form a coset of its stabilizer $G(K/F(\alpha))$. Thus

$$f(x) = \operatorname{irr}(\alpha, F)^{[K:F(\alpha)]}.$$

Since the minimal polynomial is nonconstant, this equals the minimal polynomial exactly when the exponent is 1, namely when $K = F(\alpha)$.

Exercise 53.25: A compositum corresponds to an intersection

An automorphism fixes the join $E \vee L$ if and only if it fixes both E and L: one direction follows from containment, and the other because it then fixes every rational expression in elements of the two fields. Therefore

$$G(K/(E \vee L)) = G(K/E) \cap G(K/L).$$

This is the reversal of taking the smallest field containing two given fields.

Exercise 53.26: An intersection corresponds to a generated subgroup

Put $H = \langle G(K/E), G(K/L) \rangle$. An element is fixed by H exactly when it is fixed by both generating subgroups, so

$$K_H = K_{G(K/E)} \cap K_{G(K/L)} = E \cap L.$$

The bijective Galois correspondence now gives

$$G(K/(E \cap L)) = \langle G(K/E), G(K/L) \rangle.$$

One must generate a subgroup, rather than merely take the union, which need not be a group.

54 Illustrations of Galois Theory

Exercise 54.1: Irreducibility of the eighth cyclotomic polynomial

Make the invertible substitution $x = y + 1$. Then

$$(y+1)^4 + 1 = y^4 + 4y^3 + 6y^2 + 4y + 2.$$

Eisenstein's criterion at 2 applies: every coefficient except the leading one is divisible by 2, and the constant 2 is not divisible by 4. Thus the substituted polynomial is irreducible over $\mathbb{Q}$. A factorization of $x^4 + 1$ would yield a factorization after substitution, so $x^4 + 1$ is irreducible too.

Exercise 54.2: Checking the entire octic fixed-field diagram

Put $\alpha = \sqrt[4]{2}$ and $K = \mathbb{Q}(\alpha, i)$. We have $[K : \mathbb{Q}] = 8$, since $x^4 - 2$ is Eisenstein and $\mathbb{Q}(\alpha)$ is real. To avoid relying on a picture alone, write $r(\alpha) = i\alpha, r(i) = i$ and $s(\alpha) = \alpha, s(i) = -i$. Then $r^4 = s^2 = 1$ and $srs = r^{-1}$. In the source notation $r = \rho_1$, $r^2 = \rho_2$, $s = \mu_1$, $r^2s = \mu_2$, $rs = \delta_1$, and $r^3s = \delta_2$. The proposed fields can all be verified by their fixed generators and degrees:

subgroup	proposed fixed field	degree over $\mathbb{Q}$
$\langle r^2, s\rangle = H_1$	$\mathbb{Q}(\alpha^2) = \mathbb{Q}(\sqrt{2})$	2
$\langle r\rangle = H_2$	$\mathbb{Q}(i)$	2
$\langle r^2, rs\rangle = H_3$	$\mathbb{Q}(i\alpha^2) = \mathbb{Q}(\sqrt{-2})$	2
$\langle s\rangle = H_4$	$\mathbb{Q}(\alpha)$	4
$\langle r^2s\rangle = H_5$	$\mathbb{Q}(i\alpha)$	4
$\langle r^2\rangle = H_6$	$\mathbb{Q}(\alpha^2, i)$	4
$\langle rs\rangle = H_7$	$\mathbb{Q}((1+i)\alpha)$	4
$\langle r^3s\rangle = H_8$	$\mathbb{Q}((1-i)\alpha)$	4

For each row, substitution of r, s checks that the indicated generator is fixed by the subgroup. For H_1, H_2, H_3, the generator is a nonrational quadratic radical, giving degree 2. For H_4, H_5, the fourth power is 2 and $x^4 - 2$ is irreducible. For H_6, the real quadratic field gains the nonreal i, giving degree 4. For H_7, H_8, if $\gamma = (1 \pm i)\alpha$, then $\gamma^2 = \pm 2i\sqrt{2}$ and $\gamma^4 = -8$. Its square generates $\mathbb{Q}(\sqrt{-2})$, and $\pm 2i\sqrt{2}$ is not a square there: writing $(a + b\sqrt{-2})^2 = \pm 2\sqrt{-2}$ would force $a^2 - 2b^2 = 0$ and $ab = \pm 1$, impossible with rational a, b. Thus these fields have degree 4 too. In every row this degree equals the subgroup index, proving equality with the full fixed field. The trivial subgroup fixes K; the full group fixes $\mathbb{Q}$. This verifies all ten fields and, by reversing subgroup containment, all inclusions in the diagram.

Exercise 54.3: Primitive generators and all their minimal polynomials

Use the notation of Exercise 54.2. A complete list, including top and bottom, is

field	primitive element	minimal polynomial
$\mathbb{Q}$	0	x
$\mathbb{Q}(\sqrt{2})$	$\sqrt{2}$	x^2-2
$\mathbb{Q}(i)$	i	x^2+1
$\mathbb{Q}(\sqrt{-2})$	$i\sqrt{2}$	x^2+2
$\mathbb{Q}(\alpha)$	α	x^4-2
$\mathbb{Q}(i\alpha)$	$i\alpha$	x^4-2
$\mathbb{Q}(\sqrt{2},i)$	$\sqrt{2}+i$	x^4-2x^2+9
$\mathbb{Q}((1+i)\alpha)$	$(1+i)\alpha$	x^4+8
$\mathbb{Q}((1-i)\alpha)$	$(1-i)\alpha$	x^4+8
K	$\alpha+i$	$x^8+4x^6+2x^4+28x^2+1$

The degrees and irreducibility in the first nine rows follow from Exercise 54.2 and the four distinct sign conjugates of $\sqrt{2}+i$. For the last row, the eight images of $\alpha+i$ are $i^j\alpha+\varepsilon i$ for $0\le j<4$ and $\varepsilon=\pm1$. They are all distinct: an equality with different j would express α as an element of $\mathbb{Q}(i)$, impossible because its rational degree is 4; with equal j the signs must agree. Thus this element has degree 8 and generates K. Multiplying

$$((x-i)^4-2)((x+i)^4-2)$$

gives the displayed monic polynomial, proving it is its minimal polynomial. The identical quartics in several rows do not imply identical subfields: an abstract isomorphism can send one primitive root to another.

Exercise 54.4: The fifth-root-of-unity correspondence

Let ζ be a primitive fifth root.

(a) Its powers $1,\zeta,\zeta^2,\zeta^3,\zeta^4$ are all roots of x^5-1, so $K=\mathbb{Q}(\zeta)$ is exactly its splitting field.

(b) An automorphism preserves order and the equation $\zeta^5=1$, so sends ζ to ζ^a for $a=1,2,3,4$.

(c) Φ_5 is irreducible of degree 4. Thus every listed image gives an automorphism, $\sigma_a(\zeta)=\zeta^a$. Composition multiplies exponents modulo 5. Since 2 has order 4 modulo 5, the group is $\langle\sigma_2\rangle\cong\mathbb{Z}_4$.

(d) Its only proper nontrivial subgroup is $\langle\sigma_4\rangle$, generated by complex conjugation. Its fixed field is $M=\mathbb{Q}(t)$ where $t=\zeta+\zeta^{-1}$. Dividing $1+\zeta+\cdots+\zeta^4=0$ by ζ^2 gives $t^2+t-1=0$, so $M=\mathbb{Q}(\sqrt{5})$ has degree 2 and is the full fixed field. The two diagrams are the chains

$$\begin{array}{cc} K & \langle\sigma_2\rangle \\ | & | \\ M & \langle\sigma_4\rangle \\ | & | \\ \mathbb{Q} & \{1\} \end{array}$$

with correspondence $K\leftrightarrow\{1\}$, $M\leftrightarrow\langle\sigma_4\rangle$, and $\mathbb{Q}\leftrightarrow\langle\sigma_2\rangle$. Each step has index or field degree 2.

Exercise 54.5: A fifth radical over a cyclotomic base

Put $F = \mathbb{Q}(\zeta_5)$ and $r = \sqrt[5]{2}$. Since $[F : \mathbb{Q}] = 4$ and $[\mathbb{Q}(r) : \mathbb{Q}] = 5$, the tower law applied to $F(r)$ shows $5 \mid 4[F(r) : F]$. But adjoining a root of $x^5 - 2$ gives degree at most 5, so $[F(r) : F] = 5$. The presence of ζ_5 makes all roots $r\zeta_5^j$ lie in $F(r)$, so this is the splitting field. The map $r \mapsto r\zeta_5$, fixing F, has order 5 and generates all five automorphisms. Thus the group is $\mathbb{Z}_5$.

Exercise 54.6: The seventh-root-of-unity correspondence

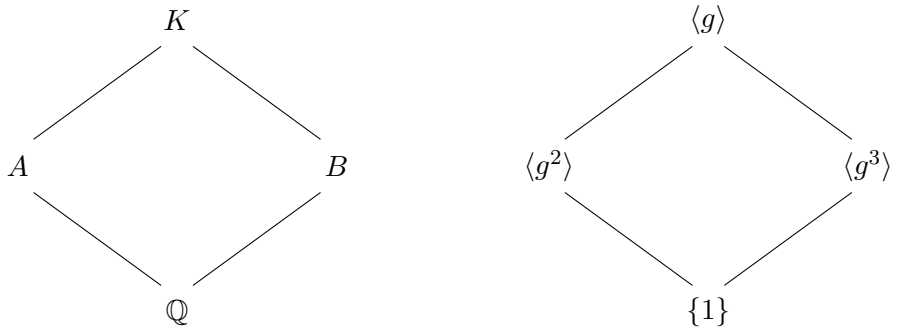
Let ζ have order 7 and $K = \mathbb{Q}(\zeta)$.

(a) Its seven powers are precisely the roots of $x^7 - 1$, so it generates the splitting field.

(b) Every automorphism sends ζ to one of ζ^a, $1 \le a \le 6$.

(c) Irreducibility of Φ_7 gives degree 6, so all six exponent maps occur. Their group is $(\mathbb{Z}/7\mathbb{Z})^\times$, cyclic of order 6 generated by 3. Write $g = \sigma_3$.

(d) The two proper nontrivial subgroups are $\langle g^2\rangle = \{\sigma_1, \sigma_2, \sigma_4\}$ of order 3 and $\langle g^3\rangle = \{\sigma_1, \sigma_6\}$ of order 2. Their fixed fields are respectively $A = \mathbb{Q}(\sqrt{-7})$ and $B = \mathbb{Q}(\zeta + \zeta^{-1})$. For the first, put $\eta = \zeta + \zeta^2 + \zeta^4$. The other three powers sum to $\bar\eta$, and direct multiplication gives $\eta + \bar\eta = -1$, $\eta\bar\eta = 2$. Thus η satisfies $x^2 + x + 2$, has degree 2, and generates A. The subgroup fixes this sum, so it fixes exactly A. For B, complex conjugation fixes $t = \zeta + \zeta^{-1}$ and ζ satisfies $X^2 - tX + 1$; since B is real and ζ nonreal, $[K : B] = 2$ and $[B : \mathbb{Q}] = 3$. For a check, $t^3 + t^2 - 2t - 1 = 0$ follows by dividing $\Phi_7(\zeta) = 0$ by ζ^3. The field and group diagrams are diamonds:



The correspondence reverses order: $A \leftrightarrow \langle g^2\rangle$, $B \leftrightarrow \langle g^3\rangle$, with top and bottom paired oppositely. There are no other intermediate fields because a cyclic group of order 6 has no other subgroups.

Exercise 54.7: Eighth roots without recomputing a table

The splitting field is $\mathbb{Q}(\zeta_8) = \mathbb{Q}(\sqrt{2}, i)$. Its group is $(\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\}$ under multiplication modulo 8. Each of its three nonidentity elements squares to 1, so it is the Klein four-group $\mathbb{Z}_2 \times \mathbb{Z}_2$.

Exercise 54.8: A complete quartic splitting-field correspondence

Find the field and its degree.

Put $a = \sqrt{2+\sqrt{5}}$, $b = i/a$, and $K = \mathbb{Q}(a, i)$. Then $b^2 = 2 - \sqrt{5}$ and $ab = i$, so the roots are $a, -a, b, -b$ and K is their splitting field. The norm of $2 + \sqrt{5}$ from $\mathbb{Q}(\sqrt{5})$ to $\mathbb{Q}$ is -1, so it is not a square there. Thus $[\mathbb{Q}(a) : \mathbb{Q}] = 4$; this field is real and gains degree 2 on adjoining i, so $[K : \mathbb{Q}] = 8$.

Describe the group.

Define

$$r(a) = b, \quad r(i) = -i, \qquad s(a) = a, \quad s(i) = -i.$$

To justify existence, conjugacy gives an embedding $\mathbb{Q}(a) \to \mathbb{Q}(b)$ sending a to b. The element i is outside $\mathbb{Q}(b)$: otherwise $a = i/b$ would also belong to that degree-four field, contradicting $[K : \mathbb{Q}] = 8$. Thus $x^2 + 1$ is still irreducible over $\mathbb{Q}(b)$, and the embedding extends with either choice $i \mapsto \pm i$. The map s is ordinary complex conjugation. This constructs the asserted automorphisms, rather than merely proposing images. They satisfy $r^4 = s^2 = 1$, $srs = r^{-1}$, and yield eight distinct maps $1, r, r^2, r^3, s, rs, r^2s, r^3s$. Thus the group is the dihedral group of order 8.

List every subgroup and its field.

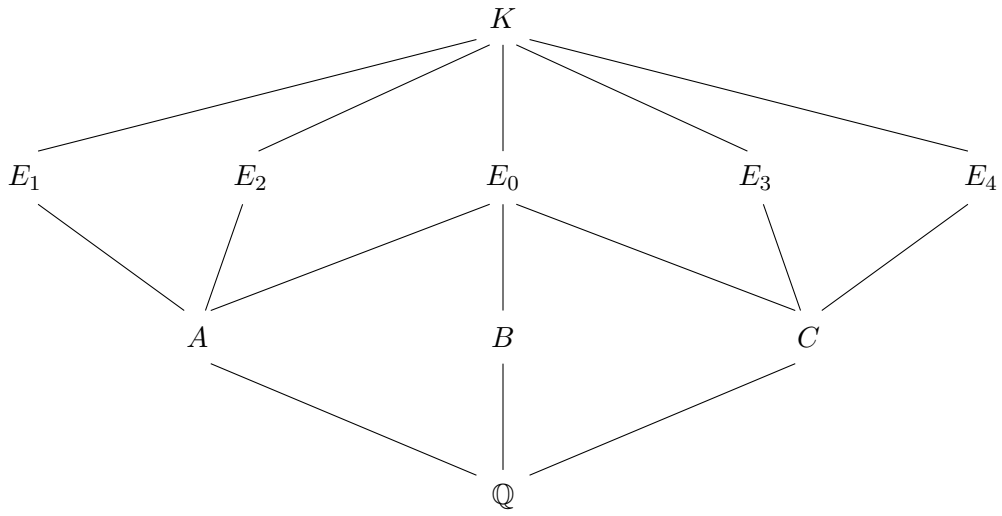
Define the field labels by this table:

subgroup	fixed field	[field : $\mathbb{Q}$]
G	$\mathbb{Q}$	1
$\langle r \rangle$	$B = \mathbb{Q}(i\sqrt{5})$	2
$\langle r^2, s \rangle$	$A = \mathbb{Q}(\sqrt{5})$	2
$\langle r^2, rs \rangle$	$C = \mathbb{Q}(i)$	2
$\langle r^2 \rangle$	$E_0 = \mathbb{Q}(\sqrt{5}, i)$	4
$\langle s \rangle$	$E_1 = \mathbb{Q}(a)$	4
$\langle r^2 s \rangle$	$E_2 = \mathbb{Q}(ia)$	4
$\langle rs \rangle$	$E_3 = \mathbb{Q}(a + b)$	4
$\langle r^3 s \rangle$	$E_4 = \mathbb{Q}(a - b)$	4
$\{1\}$	K	8

For verification, r negates both i and $\sqrt{5} = a^2 - 2$, hence fixes $i\sqrt{5}$. The subgroup $\langle r^2, s \rangle$ fixes a^2, and $\langle r^2, rs \rangle$ fixes i. The individual involutions fix their displayed generators: r^2 fixes a^2, i; s fixes a; r^2s fixes ia; rs interchanges a, b; r^3s interchanges $a, -b$. The quadratic degrees are immediate. The field $E_1 = \mathbb{Q}(a)$ has degree 4 as proved above. For E_2, the element ia satisfies $X^4 + 4X^2 - 1 = 0$. Its images under the already constructed automorphisms include $ia, -ia, ib, -ib$, four distinct numbers: $\pm ia$ are nonzero imaginary and $\pm ib = \mp 1/a$ are nonzero real. Its minimal polynomial therefore has at least four distinct roots and degree at most 4, so its degree is exactly 4. For E_0, i doubles the real quadratic degree. Finally $(a \pm b)^2 = 4 \pm 2i$, which is not a square in $\mathbb{Q}(i)$: the norm of a putative square would be the rational square 20, impossible. Hence E_3, E_4 have degree 4, with minimal polynomial $x^4 - 8x^2 + 20$. All proposed fields therefore have the required index and equal the full fixed fields.

Give all inclusions.

The following is the full field diagram. Reverse its edges and replace each field by its subgroup from the table to obtain the full subgroup diagram. In particular E_0 contains all three quadratic fields; E_1, E_2 contain only A, and E_3, E_4 contain only C.



To make the subgroup diagram explicit as well, its covering relations are

upper subgroup	immediately lower subgroups
G	$\langle r\rangle,\ \langle r^2, s\rangle,\ \langle r^2, rs\rangle$
$\langle r\rangle$	$\langle r^2\rangle$
$\langle r^2, s\rangle$	$\langle r^2\rangle, \langle s\rangle, \langle r^2 s\rangle$
$\langle r^2, rs\rangle$	$\langle r^2\rangle, \langle rs\rangle, \langle r^3 s\rangle$
each order-two subgroup	$\{1\}$

These relations and the field diagram exhibit the entire correspondence, not just its normal subgroups.

Exercise 54.9: Two symmetric-function identities

Write $s_1 = y_1 + y_2 + y_3$, $s_2 = y_1y_2 + y_1y_3 + y_2y_3$, $s_3 = y_1y_2y_3$.

(a) Expanding s_1^2 gives $y_1^2 + y_2^2 + y_3^2 + 2s_2$, so the answer is $s_1^2 - 2s_2$.

(b) Put the six displayed ratios over the denominator s_3. Their numerator is $\sum_{i\neq j} y_i^2 y_j$. On expanding s_1s_2, this sum appears together with $3y_1y_2y_3$. Thus the answer is $(s_1s_2 - 3s_3)/s_3$. The equality is an identity of rational functions, and numerical evaluation requires each original denominator nonzero.

Exercise 54.10: Polynomials of a sum and of squares

Vieta's identities for $x^3 - 4x^2 + 6x - 2$ give $s_1 = 4, s_2 = 6, s_3 = 2$ for its three roots.

(a) Their sum is 4, so the monic polynomial with that sole zero is $x - 4$.

(b) The sum of the squared roots is $s_1^2 - 2s_2 = 4$. Their pairwise products sum to $s_2^2 - 2s_1s_3 = 36 - 16 = 20$, and their product is $s_3^2 = 4$. Hence the required polynomial is $x^3 - 4x^2 + 20x - 4$. These squared roots are distinct: an opposite pair of original roots would force the third root to be 4, but the original polynomial does not vanish at 4.

Exercise 54.11: Every finite group occurs over a suitable field

Let H be finite. By Cayley's theorem embed it into S_n for some n. Take independent indeterminates $y_1, \dots, y_n$ over $\mathbb{Q}$, and put $L = \mathbb{Q}(y_1, \dots, y_n)$ and $B = \mathbb{Q}(s_1, \dots, s_n)$, using their elementary symmetric functions. The symmetric-function theorem gives $G(L/B) \cong S_n$. Regard H as a subgroup and let $F = L_H$. The Galois correspondence gives $G(L/F) = H$, and L/F is a finite normal extension. This proves existence over a suitable field F; it does not assert that every finite group occurs over $\mathbb{Q}$.

Exercise 54.12: Discriminants and even permutations

Let $\Delta = \prod_{i<j}(\alpha_i - \alpha_j)$, with roots repeated according to the multiplicities in f.

(a) $\Delta = 0$ exactly when two entries in this root list coincide, meaning f has a repeated root. Since each irreducible factor is separable, repetitions occur exactly when some irreducible factor appears at least twice. This is the stated square-divisibility criterion.

(b) An automorphism permutes the root list and sends Δ to $\pm\Delta$. Squaring removes the sign, so Δ^2 is fixed by the full Galois group and belongs to F.

(c) The equivalence needs two additional hypotheses: f must have distinct roots and $\operatorname{char} F \neq 2$. Under those hypotheses the root action embeds $G(K/F)$ into S_n, and

$$\sigma(\Delta) = \operatorname{sgn}(\sigma)\Delta.$$

This sign rule follows because interchanging two roots negates the alternating product and transpositions generate S_n. Since $\Delta \neq 0$ and $1 \neq -1$, it is fixed exactly by even permutations. Thus $G(K/F) \subseteq A_n$ if and only if every automorphism fixes Δ, equivalently $\Delta \in F$. As printed, the unrestricted assertion is false. In characteristic 2, x^2+x+1 over $\mathbb{Z}_2$ has splitting field of order 4 and group S_2, but $\Delta = \alpha - \alpha^2 = 1 \in \mathbb{Z}_2$. With repeated roots, $\Delta = 0 \in F$ regardless of the group, and the set of distinct roots may have fewer than n elements, so the stated identification with permutations of n distinct roots is not available. The preceding corrected version is the usual discriminant test.

Exercise 54.13: Algebraic integers form a ring

Let α, β satisfy monic integer polynomials of degrees m, n. Higher powers of either can be reduced using its monic relation with integer coefficients. Thus the additive group

$$M = \sum_{0 \le j < m,\ 0 \le k < n} \mathbb{Z}\alpha^j \beta^k$$

is finitely generated, contains 1, and is closed under multiplication by α and β, hence by $\alpha + \beta$, $\alpha\beta$, and $-\alpha$. It is torsion free as a subgroup of the additive group of $\mathbb{C}$, so it is a free abelian group of finite rank. For each γ among these three elements, multiplication by γ on M is represented in a $\mathbb{Z}$-basis by an integer matrix A. Its characteristic polynomial is monic with integer coefficients, and Cayley–Hamilton gives $p(A) = 0$. Applying that identity to $1 \in M$ yields $p(\gamma) = 0$ in $\mathbb{C}$. Thus $\alpha + \beta$, $\alpha\beta$, and $-\alpha$ are algebraic integers. Finally 0 and 1 satisfy x and $x - 1$. These facts verify all subring conditions.

55 Cyclotomic Extensions

Exercise 55.1: Multiplying the primitive eighth-root factors

Let ζ have order 8, so $\zeta^4 = -1$ and $\zeta^8 = 1$. The primitive roots are $\zeta, \zeta^3, \zeta^5 = -\zeta, \zeta^7 = -\zeta^3$. Pairing opposite roots gives

$$\begin{aligned}\Phi_8(x) &= (x^2 - \zeta^2)(x^2 - \zeta^6) \\ &= x^4 - (\zeta^2 + \zeta^6)x^2 + \zeta^8 = x^4 + 1,\end{aligned}$$

because $\zeta^6 = -\zeta^2$. This computes all coefficients rather than just identifying the zeros.

Exercise 55.2: The twentieth-root group as an abelian group

The group is $(\mathbb{Z}/20\mathbb{Z})^\times$. Since $20 = 4 \cdot 5$ with coprime factors, the Chinese remainder theorem gives

$$(\mathbb{Z}/20\mathbb{Z})^\times \cong (\mathbb{Z}/4\mathbb{Z})^\times \times (\mathbb{Z}/5\mathbb{Z})^\times \cong \mathbb{Z}_2 \times \mathbb{Z}_4.$$

The factors have orders 2 and 4, so the group has order 8. This is its invariant-factor classification because $2 \mid 4$.

Exercise 55.3: Three Euler totients

For $n = \prod p_j^{e_j}$, $\varphi(n) = \prod p_j^{e_j - 1}(p_j - 1)$, since a residue must not be divisible by any prime divisor of n.

(a) $60 = 2^2 \cdot 3 \cdot 5$, so $\varphi(60) = 2 \cdot 2 \cdot 4 = 16$.

(b) $1000 = 2^3 \cdot 5^3$, so $\varphi(1000) = 4 \cdot 100 = 400$.

(c) $8100 = 2^2 \cdot 3^4 \cdot 5^2$, so $\varphi(8100) = 2 \cdot 54 \cdot 20 = 2160$.

Exercise 55.4: The first thirty constructible regular polygons

The criterion is $n = 2^a$ times a product of distinct Fermat primes. Enumerating these numbers increasingly, starting at $n \geq 3$, gives

$$\begin{gathered}3, 4, 5, 6, 8, 10, 12, 15, 16, 17, \\ 20, 24, 30, 32, 34, 40, 48, 51, 60, 64, \\ 68, 80, 85, 96, 102, 120, 128, 136, 160, 170.\end{gathered}$$

For completeness of the enumeration up to 170, its only possible odd prime factors are 3,5,17; the next Fermat prime is 257. No square of an odd prime is allowed. Listing $2^a d \leq 170$ for $d \in \{1, 3, 5, 15, 17, 51, 85\}$, omitting 1 and 2 and sorting, gives exactly the displayed thirty numbers.

Exercise 55.5: The smallest constructible integral-degree angle

The answer is 3°. A regular 120-gon is constructible because $120 = 2^3 \cdot 3 \cdot 5$, so its central angle $360^\circ/120 = 3^\circ$ is constructible. A 1° angle would construct a regular 360-gon, but $360 = 2^3 \cdot 3^2 \cdot 5$ contains a square of the odd prime 3. A 2° angle would similarly construct a regular 180-gon, and $180 = 2^2 \cdot 3^2 \cdot 5$ fails the same criterion. Thus neither smaller positive integral angle is constructible.

Exercise 55.6: Twelfth roots and a group of exponent two

(a) The splitting field is $K = \mathbb{Q}(\zeta_{12})$, of degree $\varphi(12) = 4$ over $\mathbb{Q}$.

(b) Its automorphisms are the exponent maps for $1, 5, 7, 11$ modulo 12. Each exponent squares to 1 modulo 12, so every automorphism squares to the identity. A group of order 4 with no element of order 4 is $\mathbb{Z}_2 \times \mathbb{Z}_2$, giving the requested classification.

Exercise 55.7: Two cyclotomic polynomials in finite characteristic

Over $\mathbb{Z}_2$, the primitive third roots are the nonidentity roots of $x^3 - 1$, so $\Phi_3(x) = x^2 + x + 1$. Over $\mathbb{Z}_3$, the same opposite-root computation as Exercise 55.1 gives $\Phi_8(x) = x^4 + 1$. It factors there as $(x^2 + x + 2)(x^2 - x + 2)$; each quadratic has no root in $\mathbb{Z}_3$. This illustrates that a cyclotomic polynomial need not be irreducible over a finite field.

Exercise 55.8: A repeated polynomial already split over the prime field

In characteristic 3,

$$x^6 - 1 = (x^2 - 1)^3 = (x - 1)^3(x + 1)^3.$$

Its only distinct roots are $1, -1$, already in $\mathbb{Z}_3$. Therefore the splitting field is $\mathbb{Z}_3$ itself, with 3 elements. Counting the formal degree as the number of distinct roots would give the wrong answer.

Exercise 55.9: Ten statements about cyclotomic fields

(a) False. Over $\mathbb{C}$, for instance, every cyclotomic polynomial of degree greater than 1 splits into linear factors. Irreducibility here is over $\mathbb{Q}$.

(b) True. These are exactly its defining zeros.

(c) False. The group has order $\varphi(n)$; for example $n = 3$ gives order 2, not 3.

(d) True. It is isomorphic to the multiplicative group of units modulo n.

(e) True. $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$ and the extension is Galois.

(f) False. $25 = 5^2$ has a squared odd prime factor; also $\varphi(25) = 20$ is not a power of 2.

(g) True. $17 = 2^{2^2} + 1$ is a Fermat prime.

(h) True for an ordinary polygon, so prime $p \geq 3$: the criterion says an odd prime p must be Fermat. If one includes the degenerate $p = 2$ case, it must be treated separately; 2 is not a Fermat prime.

(i) False. $2^{2^5}+1 = 4294967297 = 641 \cdot 6700417$ is composite.

(j) True by definition: a Fermat prime is a prime among the numbers $2^{2^k}+1$.

Exercise 55.10: Partitioning roots by their orders

Assume the characteristic does not divide n. The derivative of $x^n - 1$ is nx^{n-1}, so all its roots in a closure are distinct. Each root has a unique multiplicative order d dividing n, and is then a primitive dth root. The root set is therefore the disjoint union of the primitive-root sets for all $d \mid n$. Multiplying their monic linear-factor products gives

$$x^n - 1 = \prod_{d|n} \Phi_d(x).$$

All the cyclotomic factors have coefficients in the prime field, hence in F, so this is an identity in $F[x]$.

Exercise 55.11: The first six cyclotomic polynomials

Use $x^n - 1 = \prod_{d|n} \Phi_d(x)$ and divide by factors from proper divisors. This gives

n	$\Phi_n(x)$
1	$x-1$
2	$x+1$
3	x^2+x+1
4	x^2+1
5	$x^4+x^3+x^2+x+1$
6	x^2-x+1

For example, $x^6 - 1 = (x-1)(x+1)(x^2+x+1)(x^2-x+1)$, so removing the factors for divisors 1,2,3 leaves the factor for 6.

Exercise 55.12: The twelfth cyclotomic polynomial

The divisors of 12 are $1, 2, 3, 4, 6, 12$. An efficient cancellation is

$$x^{12} - 1 = (x^6 - 1)(x^6 + 1), \qquad x^6 + 1 = (x^2 + 1)(x^4 - x^2 + 1).$$

The first factor already contains $\Phi_1, \Phi_2, \Phi_3, \Phi_6$, and $x^2 + 1 = \Phi_4$. Thus the remaining factor is $\Phi_{12}(x) = x^4 - x^2 + 1$.

Exercise 55.13: Negating roots of odd order

If $n > 1$ is odd and ζ has order n, then $-\zeta$ has order $2n$: its nth power is -1, and if $(-\zeta)^k = 1$, then squaring gives $n \mid k$ and the odd multiple $k = n$ does not work. Conversely, negating a primitive $2n$th root gives a primitive nth root. Thus the zeros of $\Phi_n(-x)$ are exactly those of $\Phi_{2n}(x)$. For odd $n > 1$, $\varphi(n)$ is even, since invertible residues pair as $a, -a$, so $\Phi_n(-x)$ is monic. The two monic polynomials with the same distinct roots coincide.

Exercise 55.14: The compositum for coprime root orders

Let ζ have order nm, and set $\xi = \zeta^m$, $\eta = \zeta^n$, which have orders n, m. Their fields are contained in $\mathbb{Q}(\zeta)$. Conversely choose integers u, v with $um + vn = 1$. Then $\zeta = \xi^u \eta^v \in \mathbb{Q}(\xi, \eta)$. Thus

$$\mathbb{Q}(\zeta_{nm}) = \mathbb{Q}(\zeta_n, \zeta_m),$$

which is exactly the splitting field of $(x^n - 1)(x^m - 1)$.

Exercise 55.15: The product decomposition for coprime orders

Identify the three cyclotomic Galois groups with unit groups modulo nm, n, m. Restriction to the two smaller splitting fields is the map

$$(\mathbb{Z}/nm\mathbb{Z})^\times \longrightarrow (\mathbb{Z}/n\mathbb{Z})^\times \times (\mathbb{Z}/m\mathbb{Z})^\times, \quad a \longmapsto (a \bmod n, a \bmod m).$$

It is injective because equality modulo both coprime moduli implies equality modulo their product. The Chinese remainder theorem gives a unique residue modulo nm for each pair of unit residues, and that residue is also a unit, proving surjectivity. Multiplication is preserved. Therefore the group of $x^{nm} - 1$ is the direct product of the two smaller groups, as claimed.

56 Insolvability of the Quintic

Exercise 56.1: A radical extension in characteristic two

The polynomial $x^2 + x + 1$ is irreducible over $\mathbb{Z}_2$ and its splitting field is $K = \mathrm{GF}(4)$. A square root of an element of $\mathbb{Z}_2$ cannot generate it: the only elements are 0,1, whose square roots in any extension are uniquely 0,1 in characteristic 2. Nevertheless K is a radical extension. A root α of the polynomial satisfies $\alpha^3 = 1$, and $K = \mathbb{Z}_2(\alpha)$ is obtained by adjoining this cube root of the base-field element 1. A radical need not be a square root or the real positive choice of a root.

Exercise 56.2: An even polynomial of degree eight

Yes in characteristic zero. Substitute $y = x^2$ to get the quartic $ay^4 + by^3 + cy^2 + dy + e$. Every quartic is solvable by radicals, so all four y-roots lie in a radical extension. Adjoining square roots of each of them, one after another, gives a further radical extension containing both square roots of every y-root. These are exactly the roots of the original polynomial. Thus its splitting field is contained in a radical extension.

Exercise 56.3: Ten statements about radicals and solvability

(a) True. This is the definition of solvability by radicals.

(b) True in characteristic zero, by the solvable-Galois-group criterion.

(c) True. The roots of $x^{17} - 5$ are products of a seventeenth root of 5 and powers of a seventeenth root of unity; adjoining these radicals supplies a radical extension. Its group is therefore solvable.

(d) False. Although both are transcendental, they satisfy the polynomial relation $Y^2 - X = 0$ over $\mathbb{Q}$, so they are algebraically dependent.

(e) True. The group is cyclic, hence abelian and solvable.

(f) False. $x^5 - 2$ is solved by adjoining a fifth root of 2 and a fifth root of unity.

(g) True. The quartic formulas, or solvability of subgroups of S_4, prove it.

(h) False. A real cube root of 2 has degree 3 over $\mathbb{Q}$ and cannot belong to a tower of quadratic extensions, whose degree is a power of 2.

(i) False. The cubic $x^3 - x$ has rational roots $-1, 0, 1$, requiring no extension at all.

(j) True. A subnormal series with abelian quotients characterizes solvable groups, linking the group criterion to radical towers.

Exercise 56.4: The quadratic formula over a field

Multiplying $ax^2 + bx + c = 0$ by $4a$ and completing the square gives $(2ax + b)^2 = b^2 - 4ac$. Since $a \neq 0$ and characteristic is not 2, $2a$ is invertible. With $\delta^2 = b^2 - 4ac$, the roots are $(-b + \delta)/(2a)$ and $(-b - \delta)/(2a)$, so they lie in $F(\delta)$. Conversely either root determines $\delta = 2a\alpha + b$, up to sign, so

the field generated by the roots contains δ. Thus the splitting field is $F(\sqrt{b^2-4ac})$, including the trivial-extension cases where the discriminant is already a square or zero.

Exercise 56.5: A biquadratic solved by successive radicals

Put $y = x^2$. First solve $ay^2 + by + c = 0$ by adjoining $\delta = \sqrt{b^2-4ac}$, giving $y_\pm = (-b \pm \delta)/(2a)$. Next adjoin $\sqrt{y_+}$ and $\sqrt{y_-}$. The resulting tower is radical, and contains all four roots $\pm\sqrt{y_+}, \pm\sqrt{y_-}$, with coincidences allowed. Hence the splitting field is contained in this radical extension. The characteristic assumption permits division by $2a$.

Exercise 56.6: Refining a series with abelian factors

Consider one original step $N \trianglelefteq M$ with M/N abelian, and a refined chain $N = H_0 \trianglelefteq H_1 \trianglelefteq \cdots \trianglelefteq H_k = M$. For each j, H_j/N is a subgroup of the abelian group M/N, hence is abelian. By the third isomorphism theorem,

$$H_j/H_{j-1} \cong (H_j/N)/(H_{j-1}/N),$$

which is a quotient of an abelian group and is therefore abelian. Every new step lies within some original step, so this proves the assertion for the whole refinement.

Exercise 56.7: Refining solvable factors to abelian composition factors

For a step $N \trianglelefteq M$ whose quotient M/N is solvable, choose a subnormal series of M/N with abelian factors. Lift its subgroups under the quotient map $M \to M/N$; the correspondence and isomorphism theorems give a subnormal refinement from N to M with those same abelian quotients. Do this for every original step. Because the group is finite, refine further until every nontrivial quotient is simple, obtaining a composition series. Exercise 56.6 ensures the refined quotients remain abelian. Thus all composition factors are simple abelian finite groups, hence cyclic groups of prime order.

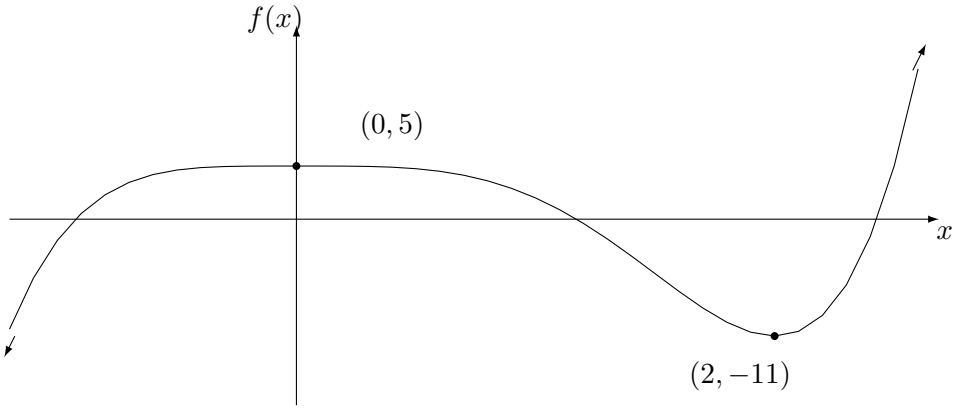
Exercise 56.8: An explicit quintic with no radical solution

(a) Label the letters by $\mathbb{Z}_5$ so the five-cycle is $c : j \mapsto j+1$. If the transposition is (u, v), its conjugates $c^k(u, v)c^{-k} = (u+k, v+k)$ all lie in H. Regard these as edges joining vertices differing by the nonzero residue $v - u$. Such edges make a connected graph, since every nonzero residue generates the additive group $\mathbb{Z}_5$. Edge transpositions of a connected graph generate all transpositions: along a path, conjugating adjacent edge swaps gives a swap of the path endpoints, by induction on its length. Hence H contains every transposition and therefore $H = S_5$.

(b) Irreducibility of the quintic makes the Galois action on its five roots transitive, so the orbit-stabilizer theorem gives $5 \mid |G|$. Cauchy's theorem gives an element of order 5, which in S_5 is a five-cycle. Complex conjugation exchanges exactly the two nonreal roots and fixes the three real ones, so its permutation is a transposition, not merely an arbitrary element of order 2. Part (a) now gives $G = S_5$.

(c) For $f(x) = 2x^5 - 5x^4 + 5$, Eisenstein at 5 proves irreducibility: 5 divides every nonleading coefficient, 25 does not divide the constant term, and 5 does not divide 2. The derivative is $f'(x) = 10x^3(x-2)$. Thus f increases on $(-\infty, 0)$, decreases on $(0, 2)$, and increases on $(2, \infty)$.

Its local maximum is $f(0) = 5 > 0$ and its local minimum is $f(2) = -11 < 0$. The odd degree with positive leading coefficient gives limits $-\infty$ and $+\infty$ at the two ends. Therefore continuity and strict monotonicity give exactly one real zero on each of these three intervals, and no others. Its remaining two roots are a nonreal conjugate pair. Here is a sketch, with the two extrema labeled and the curve drawn from actual sampled values; arrows indicate the tails.



By (b) its Galois group is S_5. This group is not solvable: its simple nonabelian normal subgroup A_5 obstructs an abelian-factor series. The solvability theorem therefore shows this quintic is not solvable by radicals over $\mathbb{Q}$.